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Identifying Colour Blindness using Computer Vision with Video Games

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of the requirements for the degree of
M.C.S. Computer Science

Declaration

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Abstract

Colour Blindness (CB), also known as Colour Vision Deficiency (CVD), can impact various aspects of daily life, such as digital interactions, career choices and education. CB is a condition which limits a person's ability to recognise and differentiate between different colours. Approximately one in twelve men and one in two hundred women globally are affected by this. There are three different forms of the condition which can affect recognition of some colours (red-green deficiency and blue-yellow deficiency) or all colours (total colour deficiency), with the most prevalent being red-green colour blindness. The idea for this dissertation stemmed from my personal experiences with CB. Current testing methods for colour blindness are often static and may not effectively engage younger audiences, prompting the development of more interactive solutions.

Within this dissertation, I explore current screening tools and testing methods, alongside current aids such as colour correction techniques (daltonization) for video games and groundbreaking innovations in this field. I also investigate the importance of CB awareness and its prevalence in society and the need for improved detection methods. With the increasing integration of technology into medical diagnostics leading to a growing interest in gamified testing methods - where interactive and immersive environments can be used to assess vision and other capabilities in a more engaging and natural manner- there is an obvious gap to be filled.

With the colour blind screening tool I developed, I propose a novel approach for identifying colour blindness using computer vision that incorporates hidden colour blind testing within retro video games, namely Pong, Space Invaders and Tetris.

After performing initial testing with adult students of Trinity College, my screening tool yielded a 100% success rate at determining colour blindness overall (separating CB participants from non CB participants), with a 100% success rate at determining CB type. These results indicate a broader use case for my colour blind screening games, especially in early childhood education, healthcare and for use by optometrists. Its proven ease of use and accuracy, along with its novel approach, make it a very promising avenue for future screening.

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1 Introduction

1.1 Motivation

Colour blindness (CB) affects a significant portion of the global population—approximately one in twelve men and one in two hundred women—yet it remains widely under diagnosed and frequently misunderstood. Although the condition has implications across many aspects of daily life, including education, employment, and digital interaction, traditional screening methods have seen little innovation in recent years. Tools such as the Ishihara plates, lantern tests, and panel-based systems are effective in clinical settings but are often rigid and lacking in user engagement, particularly for younger individuals. Their static and high-stakes nature can place users under pressure, potentially affecting the reliability of results and deterring early detection.

This project is motivated by both personal experience with CB and a broader desire to improve how the condition is identified and understood. As technology continues to play an increasingly central role in healthcare and education, there is a growing opportunity to reimagine diagnostic tools in ways that are more accessible, intuitive, and user-friendly. Gamification—particularly through retro-style video games—offers a unique means of embedding diagnostic elements in a format that is familiar, enjoyable, and discreet. Such an approach allows for the assessment of colour perception in a low-pressure environment, reducing test anxiety and improving user engagement.

By integrating CB screening mechanisms into game-based experiences, this dissertation aims to address key limitations in current testing methods while aligning with contemporary trends in digital health and inclusive design. The proposed tool not only supports early identification of CB but also contributes to awareness-raising efforts by normalising the condition within an everyday digital context. In doing so, it seeks to create a more inclusive and empathetic approach to vision testing—one that reflects the needs and expectations of a modern, diverse user base.

1.1.1 How human vision perceives colour

According to Pantone [1], colour perception arises from the brain interpreting different wavelengths of light, which are reflected by objects and detected by photoreceptors in the eye.

Human colour perception results from the complex interaction between the eyes and the brain. Light reflected off objects enters the eye and is focused onto the retina, where specialised photoreceptor cells detect it. The retina contains two primary types of photoreceptor cells: rods and cones. Rods are highly sensitive to light intensity and facilitate vision in low-light conditions, but they do not detect colour. Cones, in contrast, are active in well-lit environments and are responsible for colour perception, enabling us to distinguish between different hues based on the wavelength of light.

Each object reflects specific wavelengths of light while absorbing others. The wavelengths that are reflected determine the colour that we perceive. For instance, a tomato appears red because it reflects red wavelengths and absorbs most others. Conversely, an object that appears white reflects nearly all visible wavelengths, whereas a black object absorbs them.

The brain interprets the signals sent from the photoreceptors, resulting in the perception of colour. This process is fundamental to how we experience the world and plays a significant role in visual design, communication, and everyday recognition tasks.

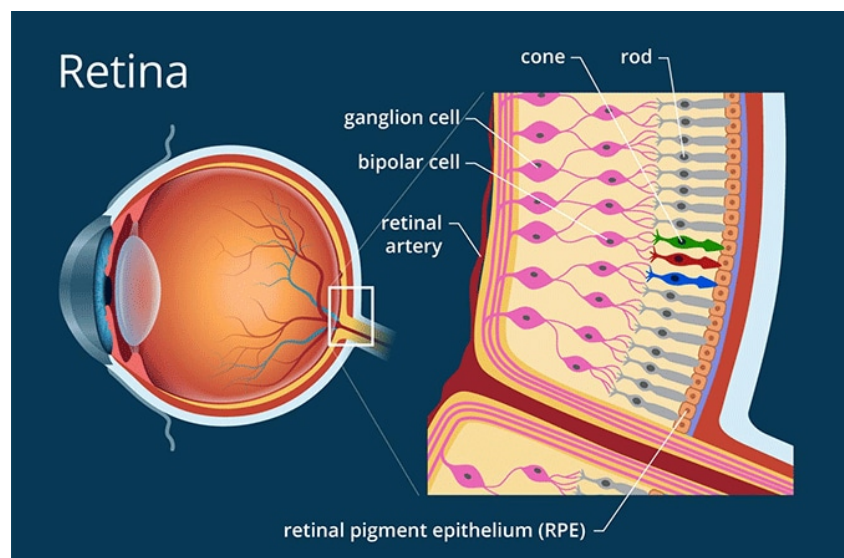


Figure 1.1: Illustration of the human retina showing the distribution of photoreceptor cells.

1.1.2 The role of cones

According to the National Institutes of Health (NIH) [2], humans possess three distinct types of cone cells, each sensitive to a specific range of wavelengths corresponding to red, green, or blue light. By combining and comparing the signals from these three cone types, the brain is able to interpret a vast spectrum of colours.

While many animals, such as mice, have only two types of cone cells (typically sensitive to green and ultraviolet or blue light), humans are trichromatic. This trichromatic system allows for greater colour discrimination and richer visual detail. Unlike in some mammals where cone types are spatially segregated within the retina, human cone cells are more uniformly distributed, allowing for more effective colour comparison and integration at the level of the brain.

Recent research has also explored the possibility of rods—typically associated with low-light vision—contributing to colour processing under certain conditions. While this is more evident in animal models such as mice, it highlights the complex interplay between different retinal cells in shaping visual perception. Nonetheless, cones remain the primary mechanism for colour detection in humans, forming the foundation for most colour-based visual experiences.

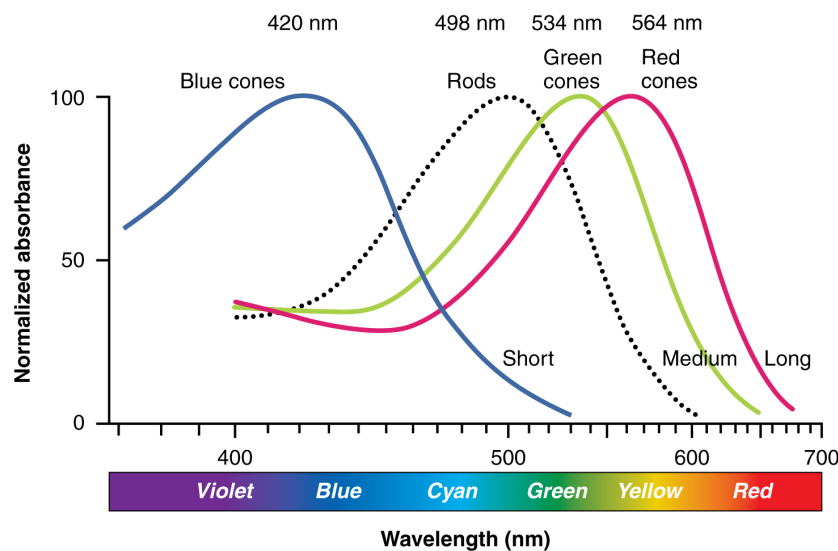


Figure 1.2: Illustration of the spectrum of light visible to the human eye

As shown in Figure 1.2, each cone cell is sensitive to a specific part of the visible spectrum.

1.1.3 The Visible Light Spectrum

The visible light spectrum refers to the portion of the electromagnetic spectrum that can be detected by the human eye [3]. It encompasses wavelengths ranging approximately from 400 to 700 nanometres. Within this range, light behaves as a wave, and each wavelength corresponds to a specific colour. Shorter wavelengths (around 380–450 nm) are perceived as violet, while longer wavelengths (up to 750 nm) appear red.

The electromagnetic spectrum as a whole includes many other types of radiation not visible to the human eye, such as gamma rays, X-rays, ultraviolet (UV), infrared (IR), microwaves, and

radio waves. Visible light occupies only a small fraction of this spectrum, positioned between ultraviolet and infrared radiation.

Each visible colour is associated with a particular wavelength. As the wavelength decreases from red to violet, the frequency and energy of the light increase. This progression is what gives rise to the familiar sequence of colours in a rainbow: red, orange, yellow, green, blue, indigo, and violet.

Light's colour arises from how different wavelengths interact with matter. When light passes through or reflects off objects, some wavelengths are absorbed while others are reflected. The reflected wavelengths are what our eyes perceive as colour.

Understanding the visible light spectrum is fundamental not only to colour vision but also to various applications in science, medicine, design, and lighting technologies.

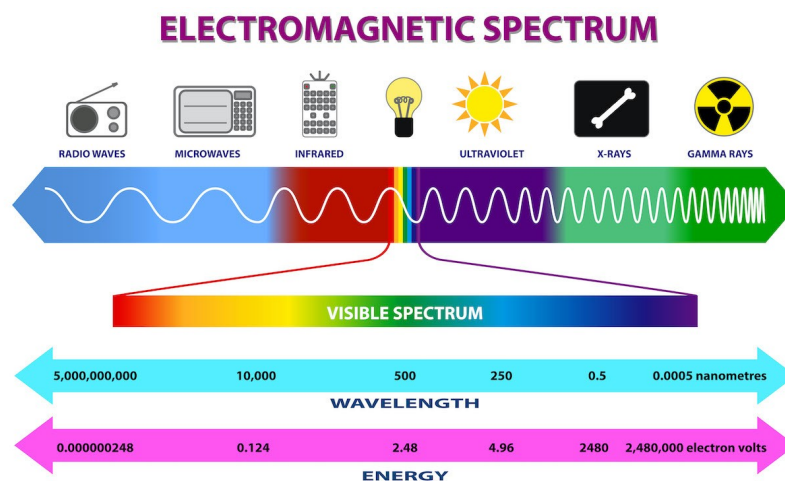


Figure 1.3: Illustration of the electromagnetic spectrum.

1.1.4 The Trichromatic Theory (Young-Helmholtz Theory)

The trichromatic theory of colour vision, also known as the Young-Helmholtz theory, posits that the human retina contains three types of photoreceptor cells, or cones, each sensitive to a specific range of wavelengths corresponding to red, green, and blue light [4]. This theory was first proposed by Thomas Young in 1802 and later expanded upon by Hermann von Helmholtz in the mid-19th century.

According to this theory, the perception of colour arises from the combined stimulation of these three types of cones. When light enters the eye, it stimulates these cones to varying degrees, and the brain interprets these signals to produce the experience of colour. For example, when both the red-sensitive and green-sensitive cones are stimulated simultaneously, the brain perceives the colour yellow.

This model explains how humans can perceive a wide spectrum of colours through the additive

mixing of red, green, and blue light. It also underpins technologies such as colour displays and digital imaging, which utilise combinations of these primary colours to reproduce a broad range of hues.

While the trichromatic theory effectively explains colour perception at the photoreceptor level, it does not account for certain visual phenomena, such as the inability to perceive certain colour combinations (e.g., reddish-green) and the occurrence of afterimages. These aspects are better explained by the opponent-process theory, which suggests that colour perception is controlled by opposing neural processes for pairs of colours: red–green, blue–yellow, and black–white.

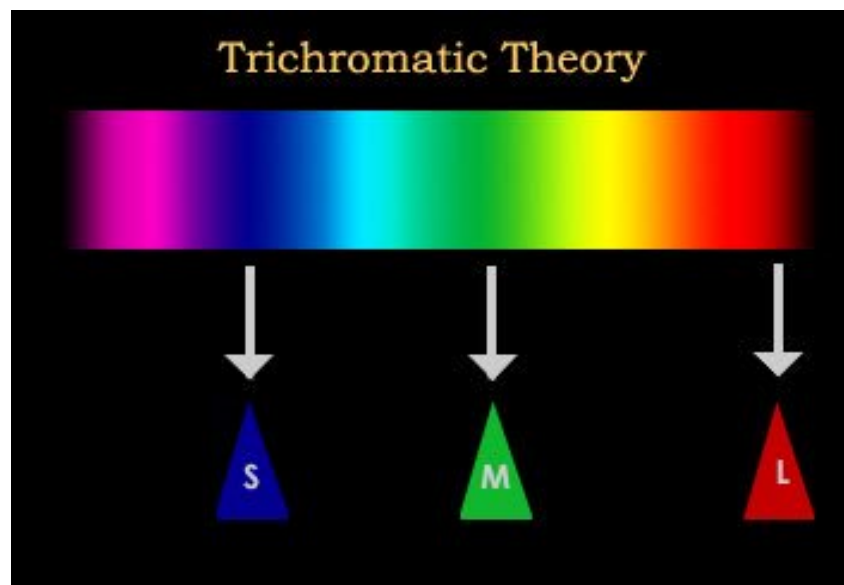


Figure 1.4: Illustration of the Trichromatic Theory of Colour Vision.

1.1.5 Opponent-Process Theory (Hering's Theory)

The opponent-process theory, proposed by Ewald Hering in 1878, offers an explanation for human colour perception by suggesting that our visual system interprets colour through three pairs of opposing channels: red–green, blue–yellow, and black–white [5]. This model posits that certain colours cannot be perceived simultaneously; for instance, one cannot visualise a reddish-green or a bluish-yellow hue. This is because the activation of one colour in a pair inhibits the perception of its opposing counterpart. For example, when the red channel is active, the green channel is suppressed, and vice versa.

This theory accounts for various visual phenomena, including negative afterimages. An afterimage is a visual illusion that occurs when the eyes are exposed to a bright colour for an extended period, causing photoreceptors in the retina to become temporarily desensitised. When the gaze is then shifted to a neutral background, the viewer perceives the complementary colour. For example, staring at a red image for some time may result in a green afterimage.

upon looking away. This is explained by the opponent-process model, where overstimulation of one side of a colour pair (e.g. red) leads to a rebound effect, triggering the perception of the opposing colour (e.g. green).

While the opponent-process theory effectively explains aspects of colour perception that the trichromatic theory does not, such as the mutual exclusivity of certain colour combinations and the occurrence of afterimages, it is now understood that both theories describe different stages of visual processing. The trichromatic theory applies to the initial stage, where the three types of cone photoreceptors detect light wavelengths. In contrast, the opponent-process theory pertains to a subsequent stage, where neural mechanisms process these signals to produce the experience of colour.

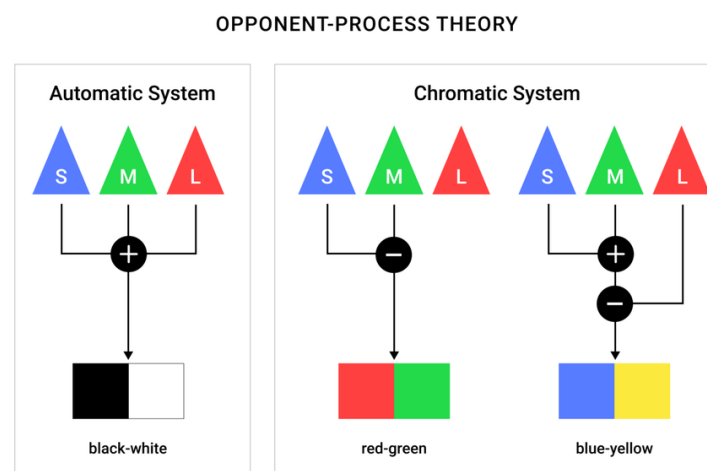


Figure 1.5: Illustration of the Opponent-Process Theory.

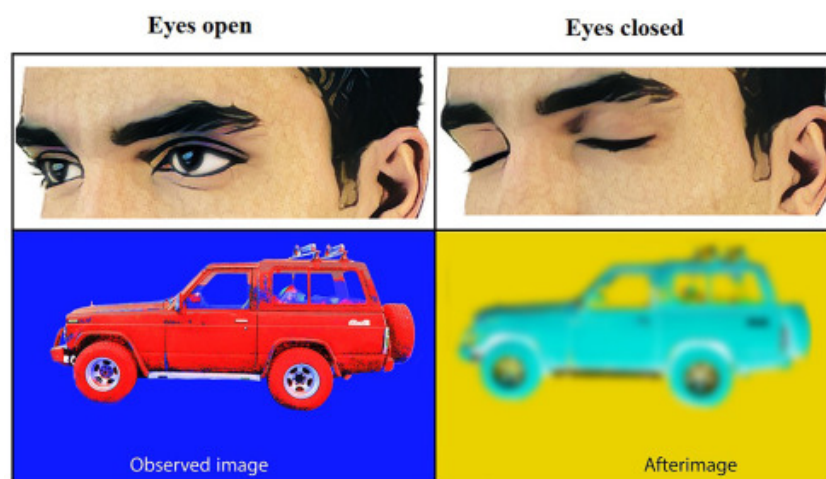


Figure 1.6: Example of an afterimage.

1.1.6 Personal Experience with Colour Blindness

The idea for this dissertation stemmed from my own personal experience of having colour blindness. From my early childhood there were some subtle signs that I had the condition. I was never able to identify or differentiate colours as easily as other children. When I looked at an object, the colour would not initially register with me, and when prompted to identify the colour, I would normally guess first and then confess that I did not know the name of the colour if my guess was wrong. When I was younger, I used to get headaches often while playing the FIFA games as the green colour of the grass would put strain on my eyes, especially if there were also red or green jerseys being worn, which I could not differentiate. As time went on, I learnt to recognise patterns in colours for certain objects. However, if I look at a bush or a tree with many different shades of red, green and brown, I see a very limited number of the shades. In a rainbow, I only see three colours maximum.

I was diagnosed with the Deuteranopia type of colour blindness, which is the red-green deficiency, when I was 11 years old. The optometrist used the Ishihara plates colour blind test to diagnose me. My primary school teachers realised indications that I might be colour blind from my taking of the Primary School Drumcondra Maths and English tests [6]. There was a section whereby a child had to match up colours, and I always found this section extremely difficult.

My diagnosis sparked a growing curiosity about how colour blindness works, why certain colours are harder for me to distinguish, and how others perceive colour differently. Over time, it became evident that this condition, although manageable, could present challenges in specific contexts — such as selecting colours for data visualisations, reading colour-coded instructions, or interpreting visual cues in games and digital media. Colour blindness also proves to be an inconvenience for myself and many others with the condition in simple everyday things like knowing what goes in each recycling bin by the colours, the important issue of being able to tell if meat is raw or cooked, or reading menus or books with yellow text on white paper.

This personal connection to colour blindness has not only influenced my academic interests but also motivated this dissertation project. By developing tools and interactive systems to identify colour blindness, I hope to create a solution that is both inclusive and subtle — one that doesn't isolate the user but instead works naturally within everyday interactions, such as through gameplay. The goal is to help others identify their own colour perception profiles early while contributing to broader awareness of how differently people may see the world.

1.2 Research Question

The main research question that will be examined throughout this dissertation is:
To Identify Colour Blindness using Computer Vision with Video Games.

1.3 Research Aims

By eliminating the usual stress that children face when being tested for a diagnosis, I aim to make the screening process for colour blindness more effective, efficient and accessible for everyone.

I carried out initial testing with adult students in the School of Computer Science and Statistics in Trinity College to see if it worked as intended. Its main purpose would be as an early diagnosis tool for colour blindness in children.

1.4 Dissertation Outline

This dissertation explores the design, development, and evaluation of a novel React Native application that gamifies the process of colour vision screening through retro-style video games. The document is structured into seven core chapters, each contributing to a comprehensive understanding of the research aims, methodology, technical implementation, and results.

Chapter One introduces the project by outlining the background, motivation, and primary objectives. It highlights the significance of exploring an unobtrusive method for colour blindness detection through interactive gameplay and explains the rationale behind combining retro games with dynamic colour manipulation and eye-tracking technology.

Chapter Two presents the background reading that underpins the project. It includes an overview of the different types of colour blindness, conventional testing methods such as the Ishihara and Farnsworth tests, and their limitations. The chapter also explores concepts related to visual perception, colour theory, accessibility, and the role of gaze tracking in modern interfaces.

Chapter Three offers a detailed review of the existing academic literature relevant to the project. It examines prior work in areas such as gamification in health contexts, eye-tracking applications, inclusive game design, and previous attempts at digital colour vision testing. This chapter identifies gaps in current research, helping to position the project within the broader field.

Chapter Four outlines the design and implementation of the system. It discusses the software tools used, the structural design of the games, and the integration of eye-tracking and dynamic colour schemes. The chapter explains how these components work together to facilitate a seamless and adaptive testing experience that does not overtly reveal the colour vision assessment taking place.



Figure 1.7: Gameplay Example from Screening Test in *Pong*.

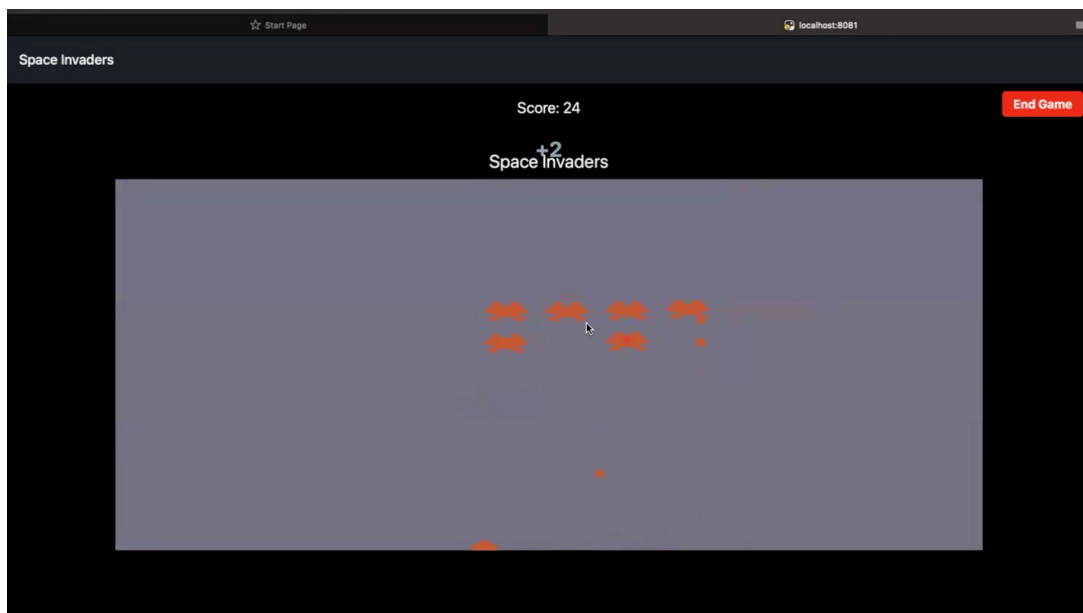


Figure 1.8: Space Invaders Protanopia Shade Pair.

Chapter Five focuses on the user testing process. It details the recruitment of participants, the testing environment, and the scenarios used to gather meaningful data. The chapter highlights key metrics such as game performance, gaze tracking patterns, and user feedback to assess both the functionality and usability of the system.

Chapter Six presents the results and evaluation. It analyses the outcomes of the user testing, exploring the system's accuracy in identifying potential colour vision deficiencies, the intuitiveness of gameplay, and how effectively the adaptive elements contributed to the overall objective. Limitations of the approach are discussed, along with any patterns or anomalies observed in the data.

Chapter Seven concludes the dissertation by summarising the key findings and reflecting on the

overall contribution of the work. It outlines opportunities for future research and development, suggesting how the system could be expanded, refined, or applied to broader use cases in healthcare, accessibility, or educational technology.

Collectively, these chapters demonstrate the feasibility of using gamified tools to passively screen for colour blindness, offering a non-invasive and engaging alternative to conventional testing approaches.

2 Background Reading

2.1 What is Colour Blindness (CB)?

Colour Blindness (CB), also known as Colour Vision Deficiency (CVD), is a condition which limits a person's ability to recognise and differentiate between different colours. Approximately one in twelve men and one in two hundred women globally are affected by this. There are three different forms of the condition which can affect recognition of some colours (red-green deficiency and blue-yellow deficiency) or all colours (total colour deficiency), with the most prevalent being red–green colour blindness.

Despite its name, this condition does not imply that affected individuals can only confuse red with green [7]. Instead, they may have difficulty distinguishing between colours that contain red or green elements. For example, they might confuse blue and purple because they cannot perceive the red component of purple. This confusion can extend across the colour spectrum, affecting the perception of reds, greens, oranges, browns, purples, pinks, greys, and even some dark shades of black. People with mild colour deficiencies learn to associate colours with certain objects and are usually able to identify colours very well. However, they are unable to appreciate colour in the same way as those with normal colour vision. People who have complete (known as total) CB often have other serious eye problems.

This condition affects approximately 8% of men and 0.5% of women globally. In the United Kingdom, this equates to around 3 million individuals, predominantly male. Worldwide, the prevalence is estimated at about 300 million people, a figure comparable to the population of the United States.

The majority of colour blindness cases are hereditary, typically passed from mother to son via the X chromosome. However, some instances result from other factors such as diseases (e.g. diabetes, multiple sclerosis), ageing, or as side effects of certain medications.

The severity of colour blindness can range from mild to severe. Notably, approximately 40% of colour blind students complete their education without being aware of their condition, and 60% of colour blind individuals experience daily challenges without fully realising the extent of their deficiency.

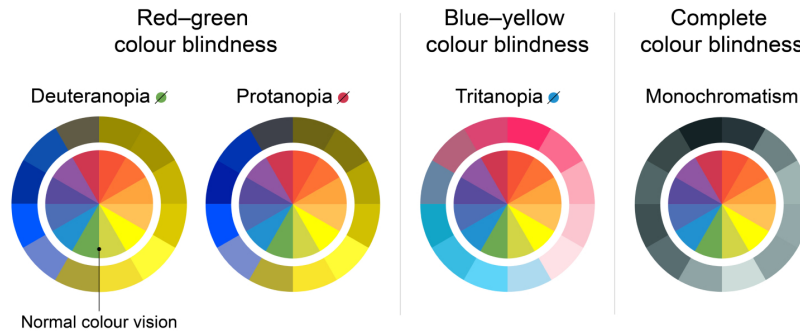


Figure 2.1: The Colour Blindness Spectrum

2.1.1 Distinction between total CB and partial CB

Total colour blindness, also known as monochromacy or achromatopsia, means a person cannot perceive any colours at all, seeing only shades of gray, while partial colour blindness (dichromacy or anomalous trichromacy) refers to difficulty distinguishing between certain colours, like red and green or blue and yellow [8].

Total Colour Blindness (Monochromacy/Achromatopsia):

A rare condition where individuals are unable to perceive any colours, seeing only black, white, and shades of gray. Achromatopsia is usually genetic, but can also be caused by brain damage. People with achromatopsia have significantly reduced visual acuity and are highly sensitive to light (photophobia). They see the world in black and white or shades of gray, similar to how someone with a grayscale monitor would see images.



Figure 2.2: Normal Vision against Monochromacy Colour Blindness

Figure 2.2 is a screenshot taken from the Colour Blind Awareness website [9].

Partial Colour Blindness (Dichromacy/Anomalous Trichromacy):

A more common condition where individuals have difficulty distinguishing between certain colours, but can still perceive some colours [10].

Dichromacy: Individuals have only two functional cone types (instead of the normal three).

Anomalous Trichromacy: Individuals have all three cone types, but their sensitivity to certain wavelengths of light is reduced, leading to difficulty distinguishing between certain colours.

2.1.2 How Colour Blindness affects perception of colours

The impact on colour perception varies depending on the type and severity of the condition [9].

Anomalous Trichromacy includes:

Protanomaly: Reduced sensitivity to red light.

Deuteranomaly: Reduced sensitivity to green light; the most prevalent form of CB.

Tritanomaly: Reduced sensitivity to blue light; extremely rare.

Individuals with protanomaly or deuteranomaly often struggle to differentiate between hues containing red and green elements, such as reds, greens, browns, and oranges. They may also confuse blues with purples, as they cannot perceive the red component in purple accurately. The degree of impairment can range from near-normal colour perception to significant difficulty in distinguishing these colours.

Dichromacy arises when one type of cone cell is absent entirely, leading to:

Protanopia: Absence of red-sensitive cones.

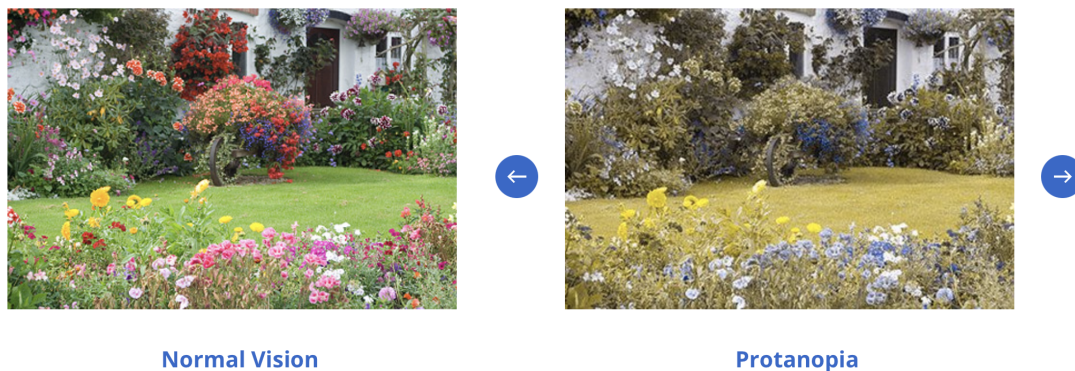


Figure 2.3: Normal Vision against Protanopia

Deuteranopia: Absence of green-sensitive cones.

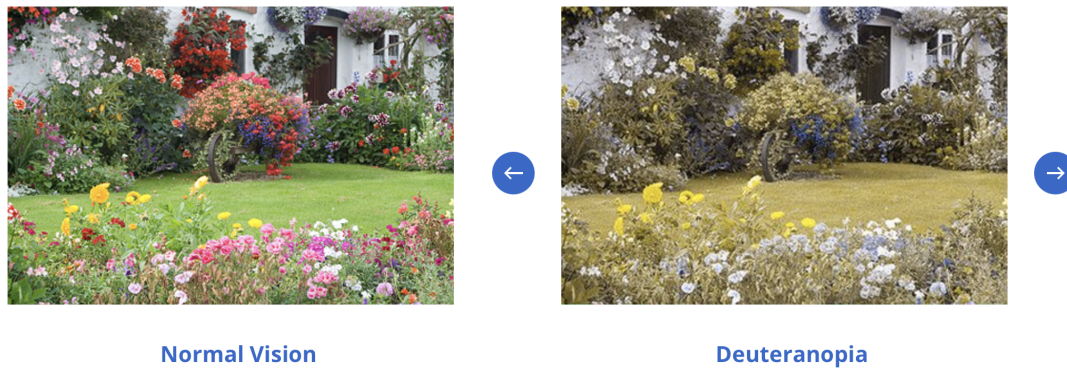


Figure 2.4: Normal Vision against Deuteranopia

Tritanopia: Absence of blue-sensitive cones.

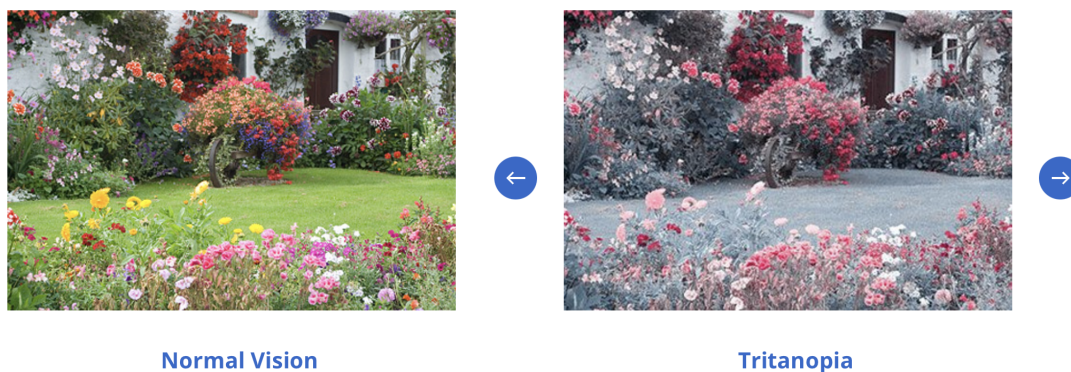


Figure 2.5: Normal Vision Against Tritanopia

Those with dichromacy perceive colours based on the two remaining functional cone types, resulting in a more pronounced inability to distinguish between certain colours. For example, individuals with protanopia or deuteranopia may see the world in shades where reds and greens are indistinguishable, affecting their perception across the colour spectrum.

Severity and Variability: The extent to which colour perception is affected varies among individuals. Some may experience mild difficulties, while others face more severe challenges in discerning colours. Additionally, lighting conditions can influence colour discrimination, with poor lighting exacerbating the difficulties faced by those with CB.

Understanding these variations is crucial for developing tools and environments that are accessible to individuals with colour vision deficiencies.

Figure 2.3, Figure 2.4 and Figure 2.5 are all screenshots taken from the 'Experience it' page of the Colour Blind Awareness website [11].

2.2 Genetic Causes

The majority of colour vision deficiencies are inherited and result from genetic mutations that affect the function or presence of cone cells in the retina [12]. The most common form, red–green colour blindness, is usually caused by a recessive gene located on the X chromosome. Since males have only one X chromosome, they are more likely to be affected if they inherit the faulty gene. Females, with two X chromosomes, are typically carriers unless both chromosomes carry the mutation, making full expression of the condition less common in women.

A mother who carries the gene has a 50% chance of passing it on to her sons. Fathers with colour blindness cannot pass the condition to their sons, but all of their daughters will be carriers. This pattern of inheritance explains the significantly higher prevalence of colour blindness in males compared to females.

The severity of inherited colour vision deficiency can vary. Some individuals experience only mild difficulties in distinguishing certain hues, while others may have more profound impairments. Importantly, inherited colour blindness is lifelong and does not typically worsen over time.

Advancements in genetic research have deepened our understanding of the specific genes involved in cone photopigment production, and how mutations in these genes disrupt normal colour perception. While there is currently no cure, this research may support future diagnostic and therapeutic developments.

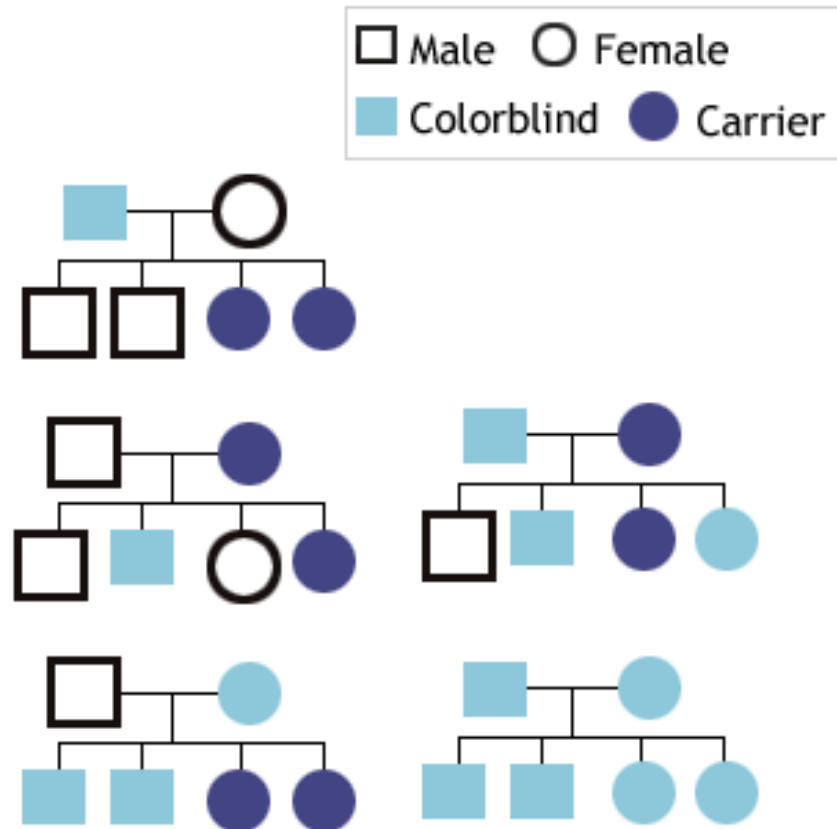


Figure 2.6: Colour Blind Genetics

2.3 Types of Colour Blindness

2.3.1 Red-Green Deficiency

Red–green colour vision deficiency is the most common form of colour blindness, affecting the ability to distinguish between colours that contain red and green elements [9]. This category includes both protan and deutan deficiencies.

Protan defects are related to anomalies or the absence of the red-sensitive (L-type) cones in the retina. Individuals with protanomaly have reduced sensitivity to red light, while those with protanopia lack red-sensitive cones altogether. As a result, reds may appear dull or brownish, and it can be difficult to differentiate red from black or dark grey.

Deutan defects involve the green-sensitive (M-type) cones. Those with deuteranomaly experience diminished sensitivity to green light and represent the most common type of red–green deficiency. In more severe cases, individuals may have deuteranopia, where green-sensitive cones are absent entirely. This leads to confusion between greens, browns, and some shades of red and orange, as well as difficulties distinguishing blues from purples.

Importantly, people with red–green colour blindness do not simply confuse red and green;

the deficiency affects a broader range of hues, particularly those that include red or green components. For example, pinks, greys, and some purples may appear similar to each other or indistinct. The overall perception of colour is altered, and colours may appear duller or less vibrant compared to how they are seen by individuals with typical colour vision.

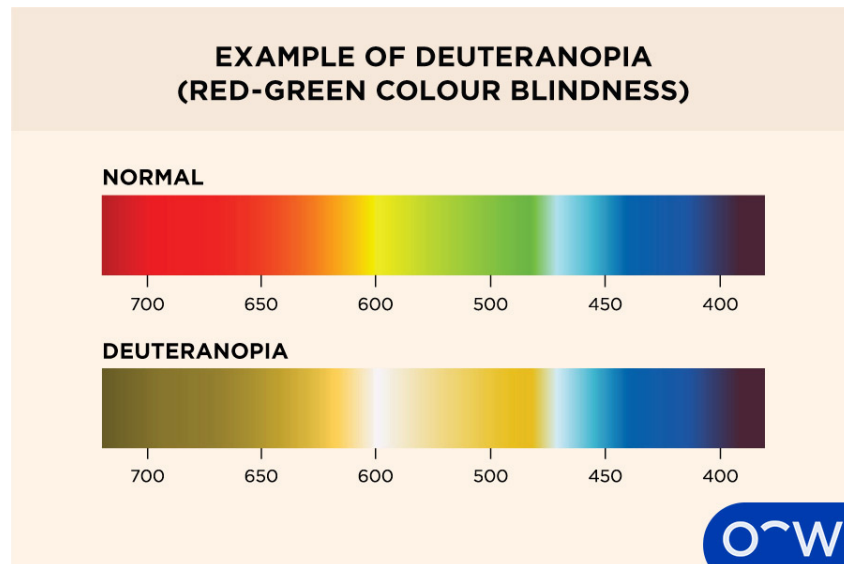


Figure 2.7: Deuteranopia CB Type Spectrum

2.3.2 Blue-Yellow Deficiency

Blue-yellow colour vision deficiency, also known as tritanomaly (when blue cone cells function abnormally) or tritanopia (when blue cone cells are absent), is a rare form of colour blindness. Individuals with this condition experience difficulty distinguishing between colours that contain blue and yellow components. Specifically, they may confuse:

Blue and green: Perceiving these hues as similar.

Violet and red: Struggling to differentiate between these colours.

Yellow and pink: Finding it challenging to tell these apart.

For those with blue-yellow deficiencies, the world often appears in shades dominated by reds, pinks, blacks, whites, greys, and turquoise. This altered perception can affect daily activities, such as interpreting colour-coded information or selecting matching clothing.

Unlike red-green colour blindness, which is commonly inherited in an X-linked recessive pattern, blue-yellow deficiencies are typically inherited differently and are not sex-linked, affecting males and females equally. However, they can also result from certain medical conditions or exposure to specific chemicals.

Understanding blue-yellow colour vision deficiency is crucial for developing inclusive designs

and accommodations, ensuring that affected individuals can navigate environments and information effectively.

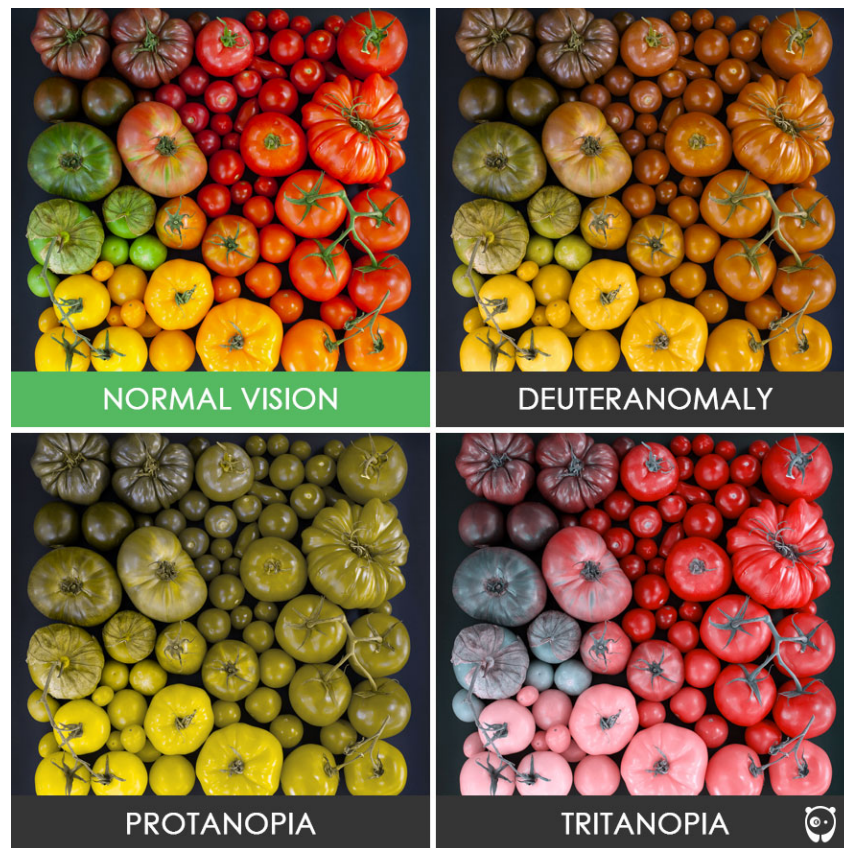


Figure 2.8: Illustration of the colours seen by the CB types against normal vision

2.3.3 A Very Rare Case

It's possible to be colour blind in one eye and have normal colour vision in the other [13]. This is called unilateral dichromacy and it's very rare.

2.4 Acquired Colour Blindness

While colour vision deficiency is often inherited, it can also be acquired due to various factors that affect the eyes or brain [14]. Unlike congenital forms, acquired colour blindness can emerge at any stage of life and may progress over time.

Chronic illnesses such as Alzheimer's disease, diabetes mellitus, glaucoma, leukaemia, liver disease, chronic alcoholism, macular degeneration, multiple sclerosis, Parkinson's disease, sickle cell anaemia, and retinitis pigmentosa have been associated with the development of colour vision deficiencies.

Neurological events, including accidents or strokes that damage the retina or specific areas

of the brain responsible for processing visual information, can lead to difficulties in colour perception.

Medications have been identified as potential contributors to acquired colour blindness. Certain antibiotics, barbiturates, anti-tuberculosis drugs, medications for high blood pressure, and treatments for nervous disorders may induce changes in colour vision as side effects.

Exposure to industrial or environmental chemicals, such as carbon monoxide, carbon disulphide, and lead-containing compounds, has been linked to the onset of colour vision defects.

Ageing also plays a role; individuals over the age of 60 may experience physical changes that affect colour perception, with research indicating a more rapid decline in colour vision after the age of 70.

It is noteworthy that most individuals with acquired colour vision deficiency retain some ability to perceive colours, often experiencing a shift similar to anomalous trichromacy. However, the severity and progression can vary, ranging from mild and stable symptoms to more severe forms that may quickly advance to monochromatism. The underlying cause often influences the extent and progression of the deficiency.

2.5 Prevalence and Demographics

2.5.1 Statistics on Global Prevalence

In the UK there are approximately 3 million colour blind people (about 4.5% of the entire population), most of whom are male [7]. Worldwide, there are estimated to be about 300 million people with colour blindness, almost the same number of people as the entire population of the USA.

Globally, approximately 8% of men and 0.5% of women experience red-green colour vision deficiency [7].

Among individuals of European descent (Caucasian populations), approximately 8% of males and 0.4% of females are affected [15]. These relatively high figures are frequently cited in this literature and are often used as baseline prevalence estimates.

In contrast, populations of East Asian descent, including Chinese and Japanese individuals, show a somewhat lower prevalence, with estimates suggesting that 4% to 6.5% of men in these groups are affected. In African populations, prevalence rates are slightly lower still, typically around 2% to 4% in males, although data is more limited and variable across regions.

These differences underscore the importance of considering ethnic background and genetic factors when estimating the impact of colour vision deficiency on global populations.

2.5.2 Difference in Prevalence among Genders

The significant disparity between male and female prevalence is attributed to the X-linked inheritance pattern of red–green colour deficiency [15]. Since males have only one X chromosome, a single copy of the defective gene will result in the condition. Females, having two X chromosomes, are more likely to be carriers unless both chromosomes carry the mutation, which is less common.

2.6 Everyday Impacts

Colour blindness can have far-reaching effects on daily life, many of which go unnoticed by individuals without the condition. From educational disadvantages to emotional challenges and safety concerns, those affected by colour blindness often experience subtle but significant limitations that impact their quality of life.

2.6.1 Challenges in Education

Educational environments frequently rely on colour-coded materials to communicate key concepts—whether in textbooks, diagrams, graphs, or interactive activities. Pupils with undiagnosed or unsupported colour blindness may struggle to distinguish between key elements, such as interpreting maps, differentiating lines in charts, or understanding coloured feedback in learning platforms. These barriers can lead to misunderstandings, reduced engagement, and even inaccurate assessments of a student’s knowledge or ability. Research suggests that a substantial number of colour blind students complete their schooling without awareness of their condition, meaning their difficulties are often misattributed to lack of effort or comprehension.

2.6.2 Difficulties in Career Paths

Certain professions place strong demands on accurate colour perception. Careers in graphic design, electrical wiring, air traffic control, aviation, medical diagnostics (e.g. pathology or radiology), and chemical analysis often involve colour-based decision-making. Individuals with colour blindness may find themselves excluded from these fields due to strict occupational vision requirements or practical limitations posed by their condition. Even in jobs without formal restrictions, misunderstandings related to colour usage—such as interpreting hazard signs, reading data visualisations, or using industry-specific software—can introduce errors or limit advancement opportunities.

2.6.3 Accessibility Challenges

One of the most frequently cited accessibility challenges relates to driving, particularly the ability to distinguish traffic lights and road signs. Although the position of traffic lights helps most colour blind drivers navigate safely, difficulties may arise in conditions where lights are misaligned, poorly visible, or viewed in peripheral vision. Red–green colour vision deficiency can cause uncertainty in recognising brake lights or interpreting red warning signs, particularly under glare or low-light conditions. While most jurisdictions permit people with colour blindness to drive, these obstacles can pose a real risk, especially for individuals who are unaware of their deficiency or receive no formal training in compensatory strategies.

2.6.4 Emotional Impacts

Beyond functional difficulties, colour vision deficiency can lead to emotional and psychological impacts, particularly when individuals feel excluded, misunderstood, or frustrated by recurring limitations. Children may experience embarrassment or bullying when they consistently "get colours wrong", and adults may feel anxiety or self-consciousness in professional or social contexts involving colour choice (e.g., selecting clothing, presenting designs, or decorating spaces). Over time, these experiences can erode confidence or reinforce a sense of difference, especially if the condition is not acknowledged or supported by those around them. Raising awareness and implementing inclusive design can help mitigate these emotional effects and promote greater understanding.

2.7 Current Diagnosis

Early diagnosis of colour blindness is essential for understanding the type and severity of an individual's condition. Several standardised tests have been developed to detect and classify colour blindness. These methods range from quick screening tools used in schools and clinical settings to highly specialised instruments and digital applications.

2.7.1 Ishihara Plates

The Ishihara Test is a widely used screening method for identifying red–green colour vision deficiency [16]. Originally developed by Dr Shinobu Ishihara in 1917, the test consists of a series of circular plates, each filled with a pattern of coloured dots. Within these dots, numbers or shapes are embedded using a colour that contrasts with the background for individuals with typical colour vision but becomes indistinguishable for those with specific colour vision deficiencies.

Each plate is carefully designed to assess whether the viewer can differentiate between certain hues. The patterns often contain numerals that individuals with normal vision can easily

identify, while those with red–green deficiencies may perceive incorrect figures or none at all. The test typically begins with plates that everyone should be able to read (to confirm understanding), followed by diagnostic and classification plates that help identify the presence and type of deficiency.

Although the Ishihara test is highly effective for detecting red–green deficiencies, it is not suitable for identifying blue–yellow colour blindness or grading the severity of the condition. Nevertheless, due to its simplicity, portability, and speed, it remains one of the most popular and reliable tools for initial screening, especially in educational, clinical, and occupational settings.



Figure 2.9: Ishihara Plates Test Examples

2.7.2 Farnsworth D-15

The Farnsworth D-15 test is a diagnostic tool designed to detect moderate to severe colour vision deficiencies and to distinguish between different types of colour blindness. Unlike simple screening tests, the D-15 offers more detailed insights into the nature of the deficiency, including whether it affects the red–green or blue–yellow spectrum.

The test consists of 15 coloured caps (or tiles), each varying subtly in hue. The participant is asked to arrange these caps in a sequence that forms a smooth gradient starting from a fixed reference colour. Individuals with normal colour vision typically arrange the caps in the correct order. However, those with colour vision deficiency often make specific patterns of errors, known as "confusion lines," that reveal the type and axis of their colour discrimination problem.

The D-15 test is often used in occupational settings where a minimum level of colour discrimination is required but not necessarily perfect colour vision. It is also employed in clinical

environments to support more comprehensive assessments. While it is more informative than the Ishihara test, it may still not detect very mild cases of colour blindness.

Both physical and digital versions of the Farnsworth D-15 test exist, though accuracy in digital formats can be influenced by screen calibration and lighting conditions.



Figure 2.10: Farnsworth D-15 Test

2.7.3 Farnsworth-Munsell 100 Hue Test

The Farnsworth-Munsell 100 Hue Test is a highly detailed diagnostic tool used to evaluate an individual's ability to perceive subtle differences in colour hues [17]. Unlike simpler screening methods, this test provides an in-depth analysis of colour discrimination across the entire visible spectrum.

The test comprises four separate trays, each containing a sequence of coloured caps with subtle gradations in hue. Each tray has two fixed endpoint caps, and the participant is asked to arrange the remaining caps in a continuous hue progression between those endpoints. The more accurately the colours are arranged, the better the person's colour discrimination.

Performance is scored based on the number and types of errors made in arranging the caps. The pattern of mistakes can reveal not only the severity of a colour vision deficiency but also the specific axis (e.g. red–green or blue–yellow) along which the individual struggles. This makes the test useful for both clinical assessment and occupational screening, particularly in industries where fine colour judgement is essential.

While the full version includes 85 caps, digital adaptations and simplified versions exist. However, accuracy in digital tests can vary due to differences in screen quality and ambient lighting.

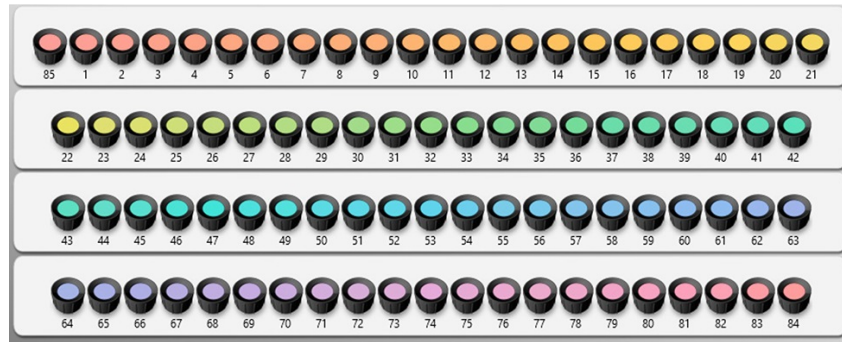


Figure 2.11: Farnsworth-Munsell 100 Hue Test

2.7.4 Lantern Test

The Lantern Test is a colour vision screening method traditionally used to assess whether individuals meet the colour perception requirements for certain occupations, particularly in fields such as aviation, maritime navigation, railways, and the military [18]. It is designed to simulate real-world conditions in which quick and accurate colour recognition is critical for safety.

The test typically presents the participant with pairs of coloured lights—usually red, green, and white—in different combinations and brightness levels. The task is to correctly identify the colours of the lights, which may be shown simultaneously or in rapid succession. This evaluates the participant's ability to distinguish these critical colours under time-constrained and variable lighting conditions.

There are several types of lantern tests in use globally, such as the Farnsworth Lantern (FALANT) used by the U.S. Navy, and the Holmes-Wright Lantern Test, commonly used in the UK and Australia. Although these tests are not designed to fully diagnose colour blindness, they are effective at determining whether an individual with a colour vision deficiency can function safely in environments where colour recognition is essential.

Digital versions of the lantern test now exist to provide preliminary screening online, although results from these tools should be interpreted with caution due to inconsistencies in display accuracy and environmental lighting.



Figure 2.12: Lantern Colour Blind Test

2.7.5 Anomaloscope (Red–Green Colour Blindness Test)

The anomaloscope is a highly specialised instrument used to diagnose and quantify red–green colour vision deficiencies with precision [19]. It is considered the gold standard for distinguishing between protan (red-deficient) and deutan (green-deficient) anomalies and for determining the severity of the deficiency.

The test involves a visual task in which the participant is asked to match a split-field image. The top half of the field displays a yellow light of adjustable brightness, while the bottom half shows a mixture of red and green light. The participant's task is to adjust the intensity of the yellow field and the ratio of red to green in the bottom field until the two halves appear to match in colour and brightness.

Individuals with normal colour vision can typically match the fields within a narrow range of red–green mixtures. However, those with red–green deficiencies require markedly different settings to achieve what they perceive as a match. The resulting values indicate both the type and degree of colour vision impairment.

Although the anomaloscope provides detailed and reliable results, its complexity, cost, and time requirements mean it is usually reserved for clinical and research settings rather than for general screening. Nevertheless, it remains a critical tool in occupational vision testing and scientific studies on colour perception.

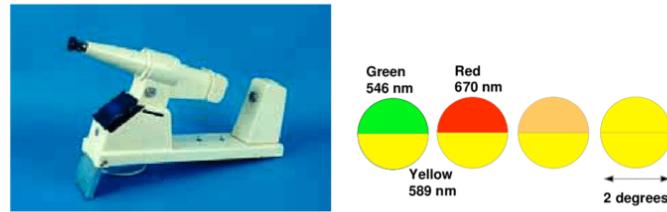


Figure 2.13: Anomaloscope test

2.7.6 Cambridge Colour Test

The Cambridge Colour Test (CCT) is a computer-based diagnostic tool designed to assess colour discrimination thresholds with a high degree of precision [20]. It is widely used in both clinical and research contexts to evaluate congenital and acquired colour vision deficiencies.

The test is based on a modified version of the Ishihara principle, presenting the subject with a series of pseudoisochromatic plates composed of small dots that form numerals. Unlike the traditional Ishihara test, however, the CCT dynamically adjusts the chromaticity of the stimuli and background using the CIE (Commission Internationale de l'Éclairage) colour space. This allows for quantitative measurement of the subject's colour discrimination thresholds along the protan (red), deutan (green), and tritan (blue) axes.

The test is administered via a calibrated monitor and uses psychophysical methods to estimate the subject's chromatic sensitivity. The results can be used to distinguish between different types of colour blindness and to monitor changes over time, for example, due to neurological disorders, drug effects, or occupational exposure.

The Cambridge Colour Test is particularly valued for its objectivity, repeatability, and adaptability across languages and cultures. It provides a comprehensive and standardised method for assessing colour vision that goes beyond the capabilities of traditional screening tools.

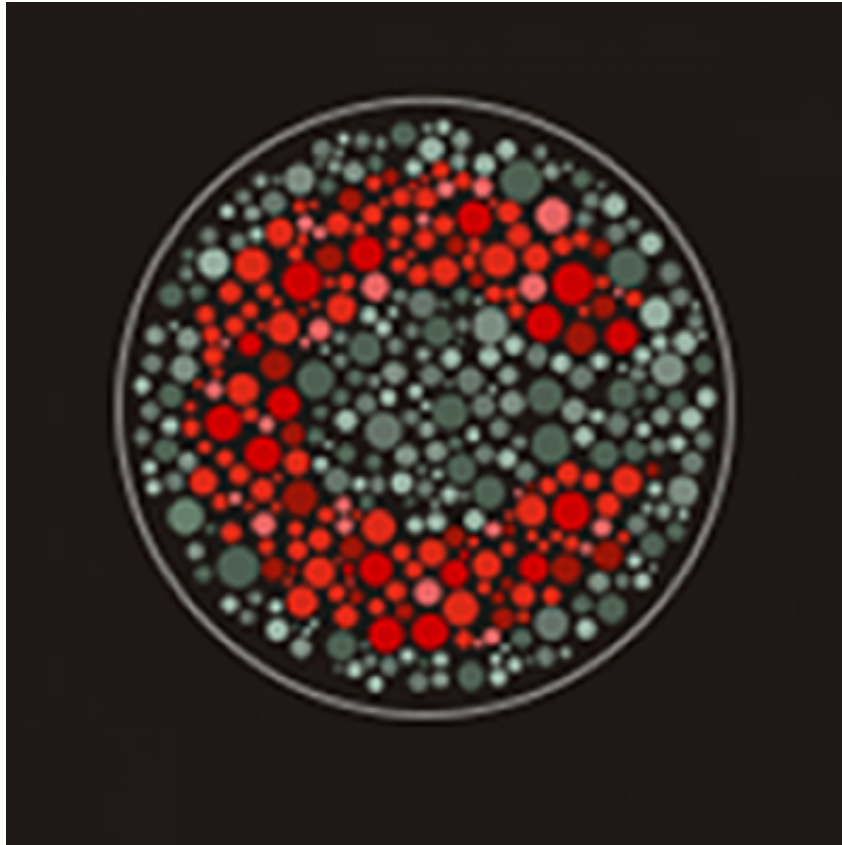


Figure 2.14: Cambridge Colour Test

2.7.7 Hardy-Rand-Rittler (HRR) Colour Vision Test

The Hardy-Rand-Rittler (HRR) Colour Vision Test is a diagnostic tool designed to identify and classify colour vision deficiencies [21]. Unlike the Ishihara test, which primarily detects red–green deficiencies, the HRR test is capable of identifying both red–green and blue–yellow (tritan) defects, making it a more comprehensive assessment of colour vision.

The test comprises a series of plates, each containing symbols such as circles, crosses, and triangles oriented in various directions. These symbols are embedded within a background of varying colours and are designed to be distinguishable by individuals with normal colour vision. Participants are required to identify the symbol and its orientation on each plate. The test begins with demonstration plates to ensure the participant understands the task, followed by diagnostic plates that assess the presence and severity of colour vision deficiencies.

One of the advantages of the HRR test is its use of symbols rather than numerals, making it accessible to individuals who are illiterate or young children who cannot read numbers. Additionally, the test's design minimises the influence of memorisation, allowing for more reliable retesting when necessary.

While digital versions of the HRR test are available, it is important to note that online colour vision tests may not yield results equivalent to the original printed versions. Factors such as

screen calibration, ambient lighting, and the limited colour gamut of digital displays can affect the accuracy of the test. Therefore, online versions should be used for preliminary screening purposes only and not for clinical diagnosis.

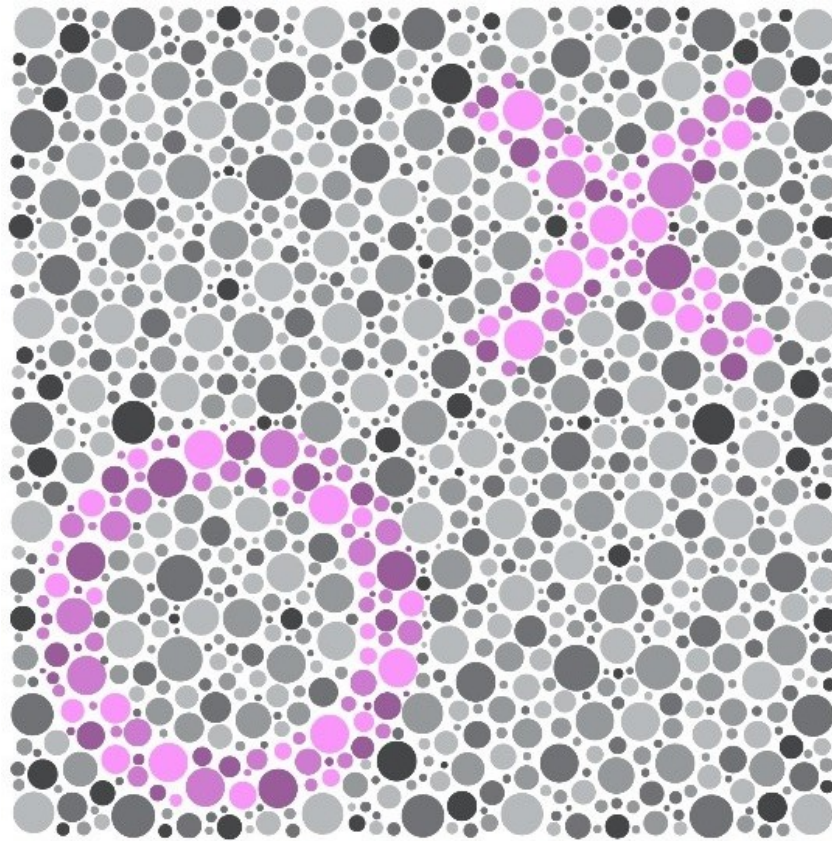


Figure 2.15: HRR Test

2.7.8 Digital and App-Based Screening Tools

Recent advances in technology have led to the development of digital platforms and mobile applications that can screen for colour vision deficiency. These tools replicate traditional tests like Ishihara plates or D-15 arrangements and often include interactive elements. Examples include EnChroma's online test, EyeQue's VisionCheck, and various mobile apps such as Color Blind Pal [22] and Color Vision Test [23,24]. While these tools increase accessibility and can raise awareness among undiagnosed individuals, their accuracy can vary depending on the display device, lighting conditions, and calibration. Therefore, they are best viewed as preliminary tools that may prompt further clinical testing.

3 Literature Review

Colour Blindness (CB) can impact various aspects of daily life, such as digital interactions, career choices and education. A recent demonstration of its awareness occurred in the sporting world when the Irish rugby team declared that they would wear white jerseys instead of their usual green in a Six Nations match to avoid clashing with Wales' red kit [25]. This was done to help supporters with CB as red and green can prove difficult to distinguish for them. Traditional colour blind tests such as the Ishihara plates, Farnsworth D-15 and Hardy-Rand-Rittler (HRR) tests have been widely used for diagnosis. However, these tests are often static, and lack adaptability to real-world scenarios, making them less effective and engaging for younger individuals. Pandey et al. critique current CB testing methods, emphasising the limitations of widely used tools like the Ishihara charts, lantern tests, and panel tests [26]. The study argues that modern occupational standards require more precise and adaptable screening protocols, especially in professions where colour discrimination is critical (e.g. aviation, maritime, and medical fields). The authors advocate for a multi-step testing protocol, starting with Ishihara charts for screening, followed by panel tests for borderline cases, and anomaloscope tests for precise classification. They further call for modernisation of lantern tests and digital adaptations to improve accuracy and accessibility.

With the increasing integration of technology into medical diagnostics, there is a growing interest in developing gamified testing methods - where interactive and immersive environments can be used to assess vision and other capabilities in a more engaging and natural manner. Recent research in accessibility and gaming suggests that games can serve as powerful diagnostic tools, capturing implicit responses to stimuli without requiring users to self-report their perceptions. Most recently, researchers from Nottingham Trent University and Kennedy Krieger Institute in the US have developed a one-minute video game to diagnose autism called Computerized Assessment of Motor Imitation (CAMI) [27]. It asks children to mimic the movements of a character on a screen, with a reported eighty per cent success rate from the study involving 180 children between seven and thirteen. This shows that there is potential for gaming to be infused further into medical diagnostics and that it is proven to yield positive results. This shift towards hidden diagnostic tools aligns with broader trends in user-centred healthcare, where passive data collection enhances both the accuracy and user experience of

medical assessments.

This chapter provides a comprehensive review of existing literature on colour blindness specific diagnostics and current accessibility innovations in video games. It explores the strengths and limitations of current CB tests, examines advancements in accessibility within gaming environments, and evaluates emerging techniques such as daltonization, augmented reality (AR) solutions, and hidden testing mechanisms. Daltonization is a colour correction technique designed to make visual content more accessible to individuals with colour vision deficiencies. It works by adjusting colours in images to reduce combinations that might be confusing for colour blind viewers [28]. By analysing previous research, this chapter identifies key gaps that inform the need for a new form of colour blindness testing - one that leverages interactive and engaging game mechanics to assess colour perception without direct user awareness of the test.

This chapter is structured as follows:

First section delves into studies investigating how CB affects gaming experiences and evaluates simulations and tools like daltonization and visual cues to enhance inclusivity. Research in this area underscores the barriers players face and offers design interventions for overcoming these challenges.

Second section explores how colour properties influence emotional engagement and immersion, revealing insights critical to crafting inclusive yet engaging game environments.

Third section focuses on groundbreaking solutions such as 3D-printed contact lenses and gene therapy, alongside practical tools like augmented reality simulations. These innovations promise transformative outcomes for managing CB.

Fourth section reviews practical solutions such as emulators and learning tools that developers can implement to address colour blindness during game design and educational efforts.

Fifth section examines the importance of colour blindness awareness in fostering inclusivity across education, work, and daily life. It highlights the challenges posed by colour differentiation and explores adaptive strategies that enhance accessibility and improve quality of life.

Sixth section synthesises the core scientific and psychological underpinnings of CB to provide a holistic understanding of its societal and personal impacts.

Seventh section identifies the need for improved detection methods and better understanding of the emotional and perceptual experiences of individuals with CB.

3.1 Colour Blindness in Gaming Environments

This section examines how CB impacts the gaming experience and the extent to which current game designs accommodate or exclude colour-blind players. A growing body of research explores the barriers CB presents in gameplay, particularly where colour is used as a primary communication or navigation tool. Studies in this area investigate simulation methods such as daltonization, alongside alternative visual cues including shapes, outlines, and text labels, offering practical interventions to enhance inclusivity. Together, these insights underscore the importance of designing with perceptual diversity in mind.

Kerac et al. examined how CB affects the player experience in online games such as Rust, Valorant, League of Legends, and Overwatch [29]. Their analysis highlighted that while some games include customisable colour filters, many still rely heavily on colour for core gameplay mechanics, limiting accessibility. The study recommends using alternative visual cues such as shapes, text labels, and outlines to supplement colour-based signals — a notion echoed in subsequent research. A notable strength of this work lies in its structured assessment of CB features across a range of titles, offering clear design recommendations. However, it focuses mainly on daltonization techniques and lacks user testing with actual CB players, omitting multimodal approaches like haptic (touch sensitive) or auditory feedback that may offer richer accessibility pathways. On the other hand, my study includes CB players and encourages verbal feedback during gameplay.

Extending the focus on developer-facing tools, Pinheiro et al. investigated how colour blindness simulation can inform accessible design practices in mobile games [30]. Their study compared the effectiveness of desktop simulations (Adobe XD's Stark plugin) with AR-based tools (SDO – Eye Disease Simulator), finding that AR simulations allowed developers to identify chromatic issues in a more immersive way. Through three empirical studies — involving developers, real CB players, and colour cue interpretation — the research revealed that simulation tools helped non-colour-blind developers identify more issues (48%) than actual CB players (32%). This highlights a gap in awareness that tools can help bridge. While the methodology is robust and practical, the study is limited by its small sample size and an overemphasis on red-green colour blindness. Similarly to Kerac et al., it reinforces the argument that colour alone should not be relied upon for conveying gameplay-critical information.

Further expanding on the limitations of colour reliance, Tillem and Gün provided a comprehensive review of colour blindness challenges in gaming [31]. They categorised prior research into design guidelines, detection methods, and technological innovations, identifying persistent gaps in understanding player performance and emotional response. Their review also cautioned against over-compensating with alternative cues such as patterns or textures, which can clutter visuals and compromise aesthetic appeal. Notably, the paper explores the use of games

like DoDo for gamified CB screening among children, aligning directly with the aims of this project — embedding subtle, diagnostic colour vision tests within retro game environments. Additionally, the study introduces the potential of AR/VR solutions such as Chroma glasses and in-game simulations, while acknowledging barriers such as cost and usability. Although the paper lacks empirical testing, its interdisciplinary approach and practical recommendations significantly enrich the current discourse.

Shifting from reviews to more targeted experimental research, Fujiwara and Nakamura explored methods to level the playing field for Deuteranopes in competitive gaming [32]. Using a Deuteranope-simulation filter, they tested colour pairs that vary more in lightness and saturation rather than hue, finding that such combinations improved contrast for colour-blind players without diminishing visibility for trichromats. Their data-driven approach offers an innovative framework for fairness in gameplay design, suggesting brightness adjustments may outperform traditional remapping strategies. This aligns with the ethos of my project, which seeks to balance accessibility and playability without separating players into mode-specific experiences. However, the study's small, non-diverse sample and lack of real-world gaming integration limit its generalisation.

Complementing this work, Lindkvist explores the effectiveness of colour, animation, and outlines in capturing visual attention in video games, with a specific focus on colour-blind accessibility [33]. Using eye-tracking technology, the study measured time to first fixation (TTFF) and total fixation duration (TFD) to assess how trichromatic and dichromatic players (deuteranopia, protanopia, tritanopia) perceive visual cues. Findings indicate that animation is the most effective method for drawing attention across all participants, while colour and outlines perform better when used together, particularly for colour-blind players. Yellow and green were most consistently effective in enhancing visibility. The study's strengths lie in its empirical methodology, quantitative approach, and focus on accessibility, offering actionable insights for improving game design for colour-blind players. However, limitations include a small sample size (20 participants), simplified experimental scenarios, and a lack of long-term analysis on gameplay impact. The research also excludes other forms of colour vision deficiency, focusing only on dichromacy.

Expanding on his earlier research in the CB space, Pinheiro, now collaborating with Andrade, Viana and Darin, investigates the effectiveness of colour blindness simulations in mobile game development, comparing desktop-based (Stark) and AR (SDO) tools for identifying accessibility issues, the same tools used in his previous study [34]. The study highlights Stark's efficiency in early-stage static design evaluation and SDO's superiority in detecting gameplay-specific accessibility issues, particularly contrast deficiencies in dynamic interactions. Findings suggest that a hybrid approach—using desktop simulations for asset design and AR simulations for real-time evaluations—improves accessibility integration throughout development. Key strengths of this study include its relevance to modern game development, practical rec-

ommendations for accessibility, and innovative use of AR for real-world testing. However, limitations include a small participant sample (10 developers), usability challenges with AR tools, and a lack of player-based testing.

3.2 Colour and Emotional Engagement in Game Design

Colour not only aids navigation and recognition in games but also plays a crucial role in shaping emotional engagement and immersion. This section explores how different hues, levels of brightness, saturation, and chromatic diversity influence players' emotional responses. Drawing on psychological models and experimental data, the reviewed studies provide a foundation for understanding how colour can be scripted to evoke specific moods, while also highlighting the need to ensure such effects remain meaningful for colour-blind players. These findings are essential for balancing accessibility with the emotional nuance that colour-rich environments provide.

Joosten et al. examine how colours influence emotional responses in video games, finding that red induces high arousal and negative emotions, while yellow evokes positive feelings [35]. Using a controlled role-playing game experiment with colour-manipulated rooms, the study measured emotional responses via the Self-Assessment Manikin (SAM). Notably, inexperienced players were more affected by colour changes than experienced players, suggesting that newer gamers rely more on visual cues for emotional engagement. The strengths of this study include its theoretical foundation in emotion-colour associations, controlled methodology, and practical insights for game design. However, limitations include a narrow participant pool (young university students), simplistic colour manipulations, and the exclusion of long-term gameplay effects. The research also does not consider colour-blind players, which may alter emotional responses due to differences in colour perception, potentially impacting the generalisability of findings.

Although the paper does not address colour blindness directly or consider how colour-blind players might perceive or respond to the colour manipulations used in the study, I believe it is worth noting that key colours used in the study, like red (anger) and green (fear) tend to appear indistinguishable for those with Protanopia and Deuteranopia CB types, potentially affecting a colour blind player's emotional response and the study's outcomes. It is possible that colour blind players might have different emotional responses or possibly none at all to the intended effects of these colours. This oversight could limit the generalisability of the findings, as players with colour blindness may not experience the same emotional cues from these colours, potentially altering their engagement or enjoyment compared to players with normal vision. Had colour-blind participants been accounted for, the study could have

provided more insights. These players might experience alternative emotions based on other visual cues—such as brightness or contrast—rather than colour alone.

Expanding on this foundation, Geslin et al. provided a more granular analysis of emotional response by deconstructing colour into its core properties — hue, brightness, saturation, and chromatic diversity — and mapping them onto emotional states using a circumplex model [36]. Using this model, their findings demonstrate that bright, saturated colours are associated with emotions such as joy and excitement, while darker, desaturated tones elicit feelings of sadness or fear. This empirical approach offers a valuable framework for emotional colour scripting in game environments, empowering designers to align colour attributes with intended mood states. However, the study’s reliance on static imagery and lack of cultural or physiological diversity, including considerations for CB players, limits its applicability to real-time, interactive gameplay. The emotional salience of colour may not translate equally across all players, particularly those with limited colour discrimination. Future studies could build on this work by exploring how colour scripting affects CB players and investigating whether other perceptual channels — such as contrast, luminance, or motion — can be leveraged to evoke similar emotional responses.

While Joosten et al. and Geslin et al. provide empirical grounding, Haapalainen offers a broader theoretical and design-focused perspective on the emotional and functional uses of colour in games [37]. Drawing on examples from *Cyberpunk 2077*, *Limbo*, *The Legend of Zelda: Breath of the Wild*, and *Overwatch*, the study analyses how colour influences world-building, narrative tone, and gameplay clarity. Haapalainen explores colour’s historical and cultural connotations, emphasizing the importance of tailoring visual design to different player demographics. A key contribution is the study’s focus on balancing aesthetic cohesion with practical accessibility, especially using alternative visual cues (e.g. patterns and textures) for colour-blind players. However, the research is predominantly qualitative and lacks direct empirical testing with players or developers. While accessibility is acknowledged, the discussion remains largely conceptual, and no specific colour-blind modes or adaptive design systems are evaluated. Incorporating user feedback and experimental validation would enhance the applicability of the study’s recommendations, which are taken into consideration in my study.

Taken together, these works illustrate the powerful role colour plays in shaping emotional experiences and functional clarity in games. However, they also underscore a significant oversight: the variability in colour perception among players, particularly those with CB. This project seeks to address this gap by designing emotion-sensitive game environments that use not only colour but also animation, contrast, and dynamic modulation to communicate mood. In doing so, it builds upon the emotional theories presented in these studies while extending their practical reach to a more inclusive player base.

3.3 Emerging Diagnostic and Assistive Technologies

Recent innovations in diagnostics and simulation tools have transformed how CB is detected, understood, and addressed. This section reviews recent advances in screening, corrective and diagnostic tools, ranging from gene therapy and 3D-printed contact lenses to augmented reality (AR) simulators. These solutions offer transformative potential by enhancing colour perception or simulating visual impairments for awareness and empathy. By assessing their benefits and limitations, this section contextualises the role of cutting-edge technologies within broader efforts to support individuals with CB in both medical and interactive settings.

One such advancement is the web-based colour vision test developed by City, University of London, which represents a notable shift toward scalable, device-agnostic CB screening [38]. Unlike traditional methods like the Ishihara charts, which require in-person administration, this tool allows remote testing through interactive digital interfaces. It is designed for accuracy across various devices, making it useful for early diagnosis, professional screening, and large-scale applications in schools and workplaces. The article highlights several strengths, including accessibility, adaptability across devices, and practical relevance for industries requiring precise colour discrimination. However, device-dependent accuracy variations and the test's limited ability to detect complex or borderline deficiencies remain challenges. Future enhancements, such as integrating machine learning for adaptive testing and refining device calibration methods, could improve precision and expand its diagnostic scope. This further indicates a gap in the market for my novel testing with colour blind games for diagnosis.

Hardware-based solutions have also gained traction. In 2018, researchers from the University of Granada evaluated the effectiveness of EnChroma glasses, which are marketed to enhance colour vision for individuals with red-green colour blindness [39]. The study involved 48 participants with colour vision deficiencies who underwent various colour vision tests, such as the Ishihara and Farnsworth-Munsell tests, both with and without the glasses. The findings indicated that while the glasses altered the appearance of colours, they did not restore normal colour vision or enable users to perceive new colours. Some participants could distinguish certain colours better when wearing the glasses, but this improvement often came at the expense of confusing other colours. The researchers concluded that although EnChroma glasses can change colour perception, they do not make a colour-blind person's vision comparable to that of individuals with normal colour.

While EnChroma glasses may not fully restore colour vision, ongoing advancements in technology are offering new and promising solutions for individuals with colour blindness. Alam et al. explore the development of 3D-printed contact lenses for CB, using poly (2-hydroxyethyl methacrylate) (pHEMA) resin combined with wavelength-selective dyes to filter problematic wavelengths [40]. The study demonstrates significant improvements in colour perception,

with CB patients achieving up to a 43% increase in correct Ishihara test identifications. The lenses were also tested for mechanical durability, water retention, and oxygen permeability, ensuring safety and comfort comparable to commercial hydrogel lenses. Strengths of this study include its innovative use of 3D printing for customised medical solutions, comprehensive material characterization, and potential for scalable, cost-effective manufacturing. However, limitations include a small sample size (only two volunteers), lack of long-term usability studies, and limited user-centred feedback on comfort and practicality. Additionally, the study primarily targets red-green and blue-yellow CB, with less emphasis on rarer colour vision impairments.

Pushing the boundaries of medical science, Mancuso et al. presented groundbreaking research in gene therapy for red-green colour blindness [41]. By injecting an L-opsin gene into the eyes of two adult squirrel monkeys, the study demonstrated that trichromatic vision could be developed later in life — challenging prior assumptions that neural circuits for colour must form during early development. The monkeys were able to differentiate colours they previously could not, and results were sustained over time. This research offers strong proof-of-concept for future human treatments, but it also underscores current limitations: the small sample size, lack of human trials, and uncertainty around subjective perception or long-term side effects. While such treatments may eventually eliminate some forms of CB, they are still years from practical, ethical deployment. In contrast, my project operates within current technological limits, using entertainment and interaction to serve as both a diagnostic and educational intervention.

Complementing these diagnostic and corrective approaches are simulation tools designed to foster empathy and raise awareness of CB. Sá et al. present the SDO: Simulador de Doenças Oculares (Eye Disease Simulator), an augmented reality (AR) application designed to simulate visual impairments in real time, including glaucoma, age-related macular degeneration (AMD), cataracts, and colour blindness (protanopia, deuteranopia, tritanopia) [42]. The app enables users to experience these conditions interactively, serving as a tool for empathy generation, medical education, and accessibility evaluation. Developed using the Unity Engine for Android, it supports low-cost head-mounted displays (HMDs) for enhanced immersion. Key strengths of this study include its innovative use of AR, broad applicability across different audiences, and cost-effective accessibility. However, limitations include a lack of rigorous user validation, a focus on static rather than dynamic visual impairments, and dependence on smartphone hardware. Additionally, there is limited documentation on how user feedback influenced development decisions.

Educational applications of CB simulation are further explored by Jeffries et al., who examined the challenges colour-blind students face in virtual learning environments (VLEs) [43]. To address this, the authors introduce the Hertfordshire Colour-Blind Emulator (HCBE), a tool that allows educators to simulate how colour-blind learners perceive educational content,

thereby encouraging reasonable adjustments as mandated by the UK's Equality Act 2010. The study's strengths lie in its timely focus on accessibility, introduction of a practical simulation tool, and clear recommendations for improving digital learning materials. However, it lacks empirical validation of HCBE's effectiveness in real-world educational settings, does not explore complementary accessibility solutions (e.g. audio descriptions), and relies heavily on qualitative rather than quantitative data. Additionally, while the paper provides practical recommendations for educators, it does not fully address the challenges of implementing these solutions, such as time constraints or lack of technical expertise.

3.4 Practical Tools for Developers and Educators

To improve accessibility for colour-blind users, a range of practical tools and educational strategies have been developed for game designers and educators. This section reviews emulator software, simulation platforms, and accessibility guidelines that allow creators to test and adapt their content for different types of colour blindness. It also considers how educational technologies can help raise awareness and foster inclusion, particularly in digital learning environments. These tools not only support compliance with accessibility standards but also encourage a more inclusive design ethos.

Nguyen advocates for the development of video games that actively incorporate colour blindness into their core mechanics, rather than simply accommodating colour-blind players [44]. The article highlights the genetic and physiological basis of CB, emphasising its prevalence (8% of men, 0.5% of women) and its impact on visual perception. Nguyen reviews existing tools like the Ishihara test and apps such as "Color Vision Helper", which assist colour-blind individuals but do not fully integrate CB into gameplay. A key contribution of this article is its call for more innovative game designs that not only address accessibility barriers but also use CB as a game mechanic to raise awareness. However, it lacks empirical evidence on how such designs impact player experience and could benefit from examples of existing games that have successfully implemented colour-blind-friendly mechanics. This aligns closely with my research on developing a colour-blind detection test within retro-style video games. While Nguyen's argument focuses on raising awareness through game design, my approach goes further by embedding subtle, dynamic colour-based tests into gameplay to both identify and adapt to a player's specific type of CB.

Katsnelson brings a similar argument to the realm of scientific communication, where colour-coded visuals often exclude viewers with CB, particularly in data-heavy STEM fields [45]. The article critiques common practices such as using red-green colour gradients in heatmaps and charts, which can lead to misinterpretation for CB individuals. To address this, Katsnelson offers a series of actionable recommendations, including the use of perceptually uniform colour palettes (e.g., viridis, cividis), redundant visual cues like patterns and textures, and software

tools such as Color Oracle and Coblis that simulate various types of colour blindness. The article is particularly strong in its practical guidance and emphasis on inclusive design. However, it focuses primarily on red-green deficiencies and lacks broader engagement with rarer CB types such as tritanopia. Moreover, while the recommendations are well-grounded, they are not supported by empirical user studies, and the piece does not deeply explore how to balance accessibility with visual aesthetics — a common tension in both game and scientific design. These insights remain relevant for my work, especially in considering how to blend visual inclusivity with retro aesthetics without compromising either function or engagement.

Together, Nguyen and Katsnelson point to a broader trend in accessibility research: moving from reactive accommodation to proactive, inclusive design. Both advocate for systems that not only support colour-blind users but actively include their experience in shaping the final product. This philosophy underpins my project's goal of integrating colour-blind detection into gameplay — not just for diagnostic purposes, but to build a more inclusive and immersive player experience that responds dynamically to individual needs.

3.5 Everyday Impacts and Awareness-Raising Approaches

Colour blindness is a widespread yet often overlooked condition that affects individuals across various aspects of life, from education and career choices to daily tasks and emotional well-being. Understanding the challenges faced by people with CB is crucial for developing inclusive solutions, particularly in fields where colour differentiation is essential. This section reviews key literature on the physiological, psychological, and occupational impacts of CB, highlighting the importance of early detection, adaptive strategies, and accessibility improvements.

Gordon offers one of the earliest comprehensive examinations of colour blindness, exploring its physiological and genetic causes alongside its impact on education and career opportunities [46]. In education, colour-blind students may struggle with subjects relying on colour-coded materials (e.g. geography maps, science diagrams, and mathematics charts). Misinterpretation of colours can lead to mislabelling these students as inattentive or careless, impacting self-esteem and learning outcomes. While some colour-blind students develop compensatory strategies, such as using position or texture cues, the paper highlights the role of educators in adapting teaching materials to improve accessibility. Gordon also outlines the career limitations faced by colour-blind individuals in aviation, medicine, navigation, electrical work, and design, where accurate colour differentiation is critical. The author advocates for early testing to guide career counselling and reduce occupational disadvantages. This echoes the aim of my study as it advocates for more effective initial CB screening for children.

Building on the theme of personal impact, Barry et al. developed and validated a CBQoL questionnaire, the first tool assessing the quality-of-life (QoL) impact of colour blindness (CB)

[47]. Their study identified three key domains of CB-related challenges: health/lifestyle (e.g. difficulties identifying food ripeness or medical cues), work/career (barriers in colour-dependent professions and workplace materials), and emotional well-being (frustration, anxiety, and self-esteem issues). With strong psychometric validation (Cronbach's $\alpha = .979$), the CBQoL highlights the daily and professional limitations imposed by CB. However, the study's online sampling and lack of CB subtype differentiation limit its generalisability. Nevertheless, the CBQoL framework provides a useful lens through which to assess the broader consequences of CB — and by extension, the potential benefits of game-based tools that can both detect and normalise the condition.

Further exploring the functional effects of CB in everyday life, Tagarelli et al. conducted a detailed study on how colour blindness impacts daily routines, occupational engagement, and driving habits [48]. Participants reported challenges with tasks such as clothing selection (23.8%), workplace performance (13.2%), and cooking (31.7%). Interestingly, while overall employment rates did not significantly differ from those of individuals with typical colour vision, CB participants were under-represented in leadership roles, suggesting possible implicit biases or self-selection due to perceived limitations. In driving, participants reported avoiding night-time travel due to difficulty perceiving road reflectors and tail-lights, although most compensated for traffic signals using positional knowledge. Accident rates were comparable to non-CB drivers, though risks may increase under specific conditions. These findings emphasise the subtle yet pervasive ways in which CB affects independence and self-confidence. In game design, similar compensatory strategies — such as relying on brightness, position, or contrast rather than hue — can be embedded to enhance inclusivity for colour-blind players. This aligns directly with my project's goal of creating retro-style games that adapt to players' visual needs without sacrificing immersion or engagement.

3.6 Prevalence and Awareness of Colour Blindness in Diverse Populations

Understanding the global prevalence and awareness of CB provides essential context for designing inclusive diagnostic and educational tools. Studies from diverse geographic and cultural contexts reveal not only variations in CB rates but also common gaps in awareness and screening, reinforcing the need for accessible, scalable solutions like the one proposed in this study.

Fakorede et al. investigate the prevalence, gender, and ethnic variations of CB among 1,191 students in tertiary institutions in Lagos, Nigeria, while also exploring its genetic basis [49]. The overall prevalence was 2.85%, with significantly higher rates in males (4.29%) compared to females (1.58%), consistent with X-linked inheritance. Deuteranomaly (1.51%) was more

common than protanomaly (1.34%). Notable differences were observed across institutions, with Yaba College of Technology (YABATECH) reporting the highest rate (4.25%), and among ethnic groups, with Yoruba students (3.57%) more affected than Igbo (2.62%) and Hausa (1.45%). Genetic analysis confirmed inheritance patterns consistent with Hardy-Weinberg equilibrium. Importantly, the study noted limited awareness of CB and its potential career restrictions in fields such as medicine, engineering, and transportation. While the dataset is robust and geographically diverse within an urban context, the study excludes rarer forms of CB such as tritanopia and total colour blindness and lacks in-depth analysis of the psychological and educational impact of CB. Nonetheless, its findings support the argument for widespread early screening and career-oriented guidance — objectives that are directly aligned with my study's goal of using retro-style games as a novel, engaging diagnostic medium.

Similarly, Kanwal et al. investigated CB prevalence and awareness among primary school children aged 5–12 in Karachi, Pakistan [50]. The study identified a surprisingly high prevalence of CB (10.8%), with males (12%) only slightly more affected than females (9.6%). Red-green deficiencies were most common, and a significant association was found between CB and night blindness ($p = 0.002$), with family history also playing a role. Strikingly, awareness levels were critically low: most children and their guardians were unaware of the condition ($p < 0.001$), highlighting a serious gap in early diagnosis and public health outreach. The study's strengths lie in its localised focus on a previously under-researched population, its clear methodology using Ishihara tests, and the novel link established between CB and night blindness. However, the small sample size, urban sampling bias, and lack of investigation into rarer CB types limit the study's generalisability. The high prevalence rate in females also suggests the possibility of sampling error or genetic misclassification. Despite these limitations, the findings emphasise the urgent need for awareness programmes and accessible screening tools, particularly in low-resource educational settings — a gap that this project seeks to address through gamified diagnostics that engage young users intuitively and effectively.

Tongbram and Naorem provide a complementary perspective by investigating the prevalence of congenital red-green colour blindness in 204 males from the Bhoksha tribe in Dehradun, India, finding a 1.47% prevalence (protanopia, deuteranopia, and achromatopsia at 0.49% each) [51]. The study supports the "relaxation of selection" hypothesis, suggesting urbanisation reduces the selective disadvantage of colour blindness. Using standardised Ishihara testing, the research highlights genetic factors like endogamy and shifting occupational roles. Strengths include its anthropological focus and methodological rigor, but limitations arise from the small, male-only sample and lack of longitudinal analysis. Future research should include female carriers, compare urban and tribal populations, and assess colour blindness' impact on education and careers. This study provides valuable genetic insights but requires broader scope for wider applicability.

3.7 Visual Attention, Perception, and Cognitive Constraints in Game Environments

Understanding how players perceive and respond to visual stimuli is essential when designing accessible, attention-sensitive game mechanics — particularly those intended to assess or adapt to vision-related conditions like CB. Research into environmental change blindness and gaze behaviour sheds light on the limitations of visual processing and highlights opportunities for enhancing gameplay through informed visual design.

Marples et al. explored environmental change blindness (ECB) in virtual environments, demonstrating that gradual visual changes often go unnoticed due to cognitive constraints such as limited Visual Short-Term Memory (VSTM) and the effects of distributed cognition [52]. Using a virtual escape room developed in Unreal Development Kit (UDK), 119 participants were exposed to gradual colour and texture changes over varying durations. Despite 52.5% reporting changes, only 4.2% could accurately identify them, with no significant impact of timing on detection rates. To further underscore the limits of visual attention, the researchers employed a “gorilla test” — a well-known in-attentional blindness demonstration — where participants failed to notice an obvious anomaly due to cognitive overload. These findings reveal the fragility of visual awareness in dynamic environments and suggest that players may miss subtle colour changes unless they are specifically designed to draw attention. This insight is particularly relevant for my study, which relies on subtle, adaptive shifts in colour to evaluate CB without disrupting gameplay flow. Designing visual cues that remain detectable even under cognitive load is essential for ensuring both diagnostic accuracy and player immersion.

Building on the interplay between visual perception and attention, Cajar et al. examine how spatial frequencies and colour influence object search in real-world scenes, focusing on central vs. peripheral vision [53]. Using a gaze-contingent eye-tracking experiment, the study revealed that central vision depends on high spatial frequencies for detailed object identification, while peripheral vision relies on low spatial frequencies for guiding attention and localising objects. Critically, the presence of colour significantly influenced search performance — particularly in peripheral vision, where its absence (in greyscale conditions) resulted in reduced accuracy and slower detection. This suggests that colour is not only important for recognition but also plays a guiding role in directing visual attention. These findings are highly applicable to my project, which integrates eye-tracking to reward gaze-based engagement and uses adaptive colour schemes to enhance visibility for players with CB. Understanding how colour and spatial frequency interact across the visual field can inform the placement and behaviour of game elements, ensuring that all players — regardless of colour perception — can successfully navigate and respond to visual cues.

Together, these studies highlight the limitations of visual processing and the importance of

carefully balancing subtlety with salience in interactive visual environments. By considering cognitive load, peripheral vision dynamics, and the role of colour in attention guidance, this project aims to create an inclusive and perceptually robust gameplay experience — one that can unobtrusively detect CB while maintaining player engagement and flow.

4 Design and Implementation

4.1 Application Overview

The aim of this project was to create an interactive and engaging web-based application that utilises gamification as a novel method for screening for colour blindness. Built using React Native with the Expo framework, the application features three retro-style arcade games—Pong, Space Invaders, and Tetris—designed to evaluate user visual performance under dynamically changing colour conditions. By utilising gamification during testing, it is intended to eliminate the feeling of being tested for the user to relieve any stress or tension that can arise from various current screening methods to give the most accurate results as possible.

A key feature of the application is the integration of dynamically cycling colour palettes, each carefully chosen to simulate colour pairings known to be difficult to distinguish for individuals with various types of colour blindness, namely protanopia, deuteranopia, and tritanopia. These palettes were informed by medically accepted guidelines and visual confusion data obtained from the Colour Blind Awareness website [54].

As users interact with the games, the application unobtrusively alters the colour scheme according to a predetermined sequence. By tracking performance metrics such as reaction time, score, and accuracy across different colour modes, the system seeks to infer whether the user may exhibit difficulty distinguishing specific colour combinations, thereby offering potential insights into their type of colour vision deficiency.

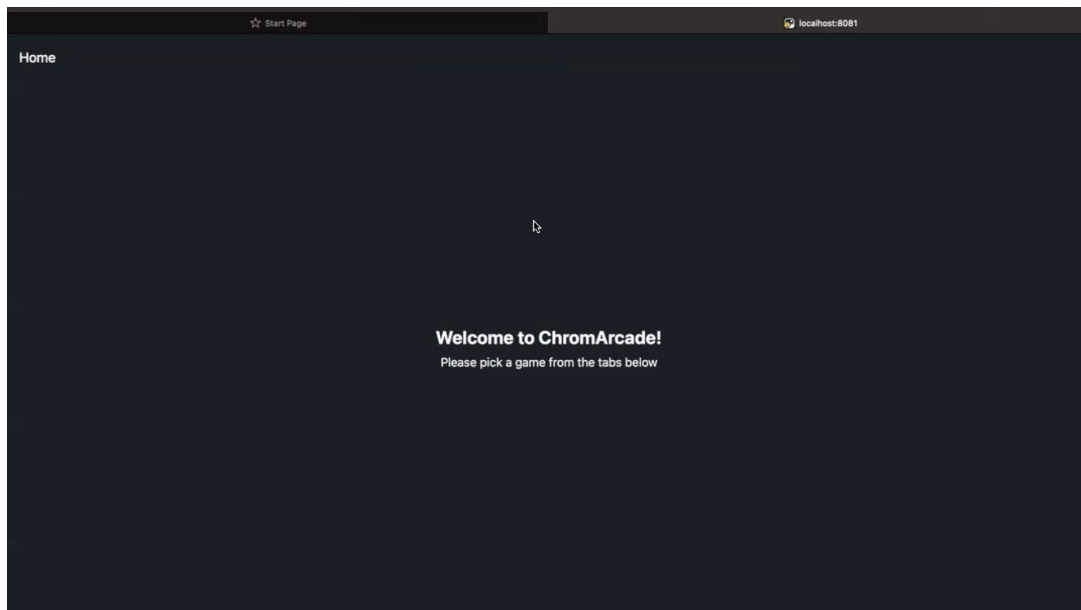


Figure 4.1: Home Page of Colour Blind Games.

4.2 Video Games Chosen

Three games were chosen for their simplicity, iconic status, and suitability for testing visual reaction under time constraints. Each game offers a unique cognitive and visual challenge that can be influenced by the ability to discern colours accurately.

4.2.1 Pong

Pong was selected due to its minimalist visual interface and strong dependence on visual tracking of a moving object. The paddle and ball serve as key elements whose visibility is essential to gameplay. This setup provides a clear context for evaluating how colour confusion impacts fast-paced object recognition and response.

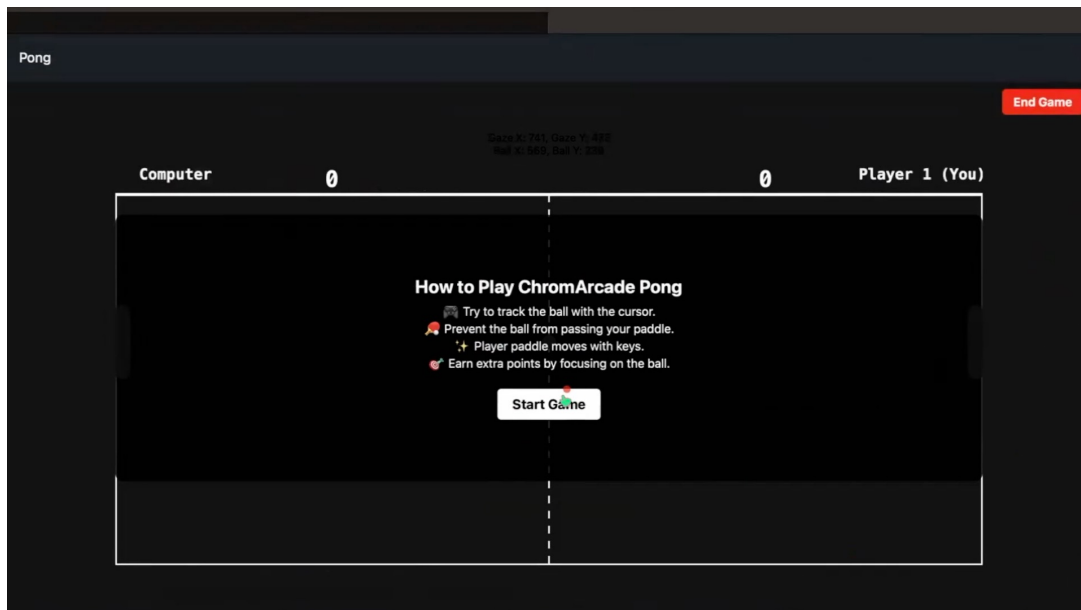


Figure 4.2: Pong In-Game Instructions.

4.2.2 Space Invaders

Space Invaders introduces greater visual complexity with multiple enemy sprites and player projectiles. Colour differentiation is critical for distinguishing between moving targets and threats. Cycling colour palettes during this game offers an effective mechanism for assessing how colour confusion affects accuracy and decision-making.

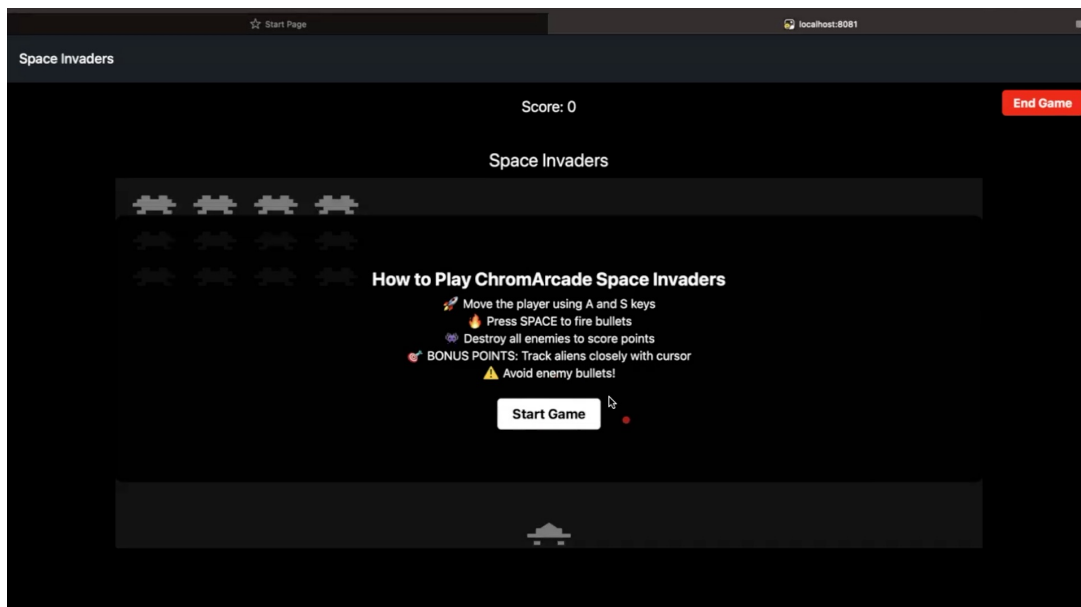


Figure 4.3: Space Invaders In-Game Instructions.

4.2.3 Tetris

Tetris was chosen due to its reliance on the identification and rotation of differently shaped and coloured blocks (tetrominoes). As colours are cycled to emulate visual confusion common in CB, players may find it increasingly difficult to differentiate between similar-shaped blocks. The game serves as an effective test of both visual processing and spatial reasoning under colour-restricted conditions.

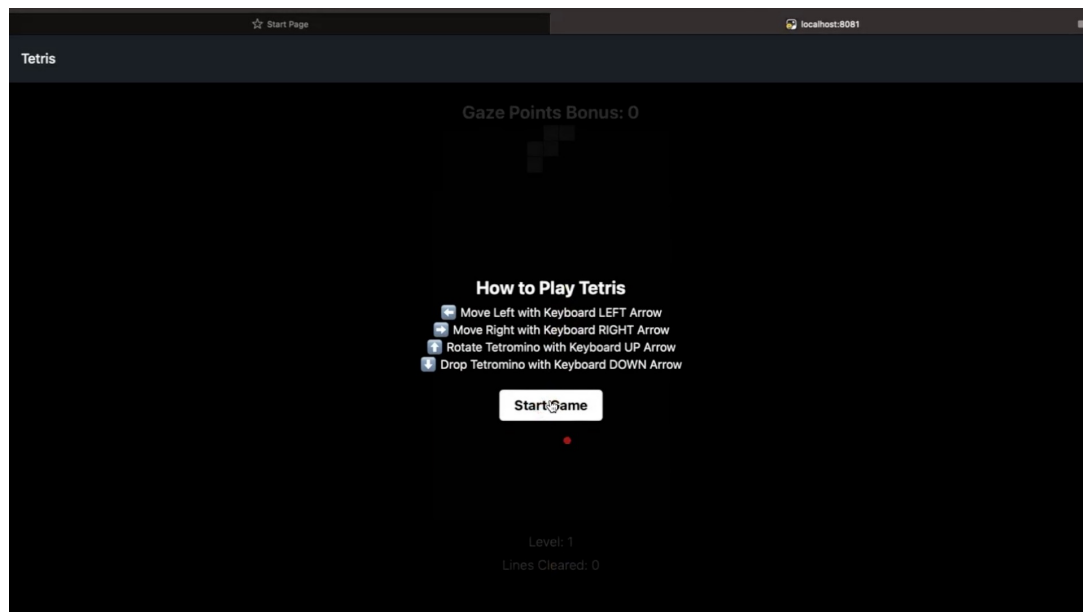


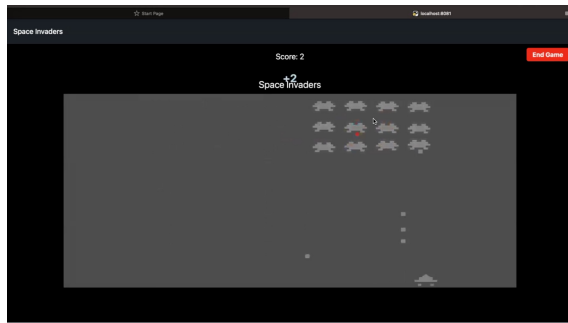
Figure 4.4: Tetris In-Game Instructions.

The most important aspect to keep in mind in relation to these games is that they would be simple for anyone, especially a child, to pick up and play, which introduces the accessibility aspect.

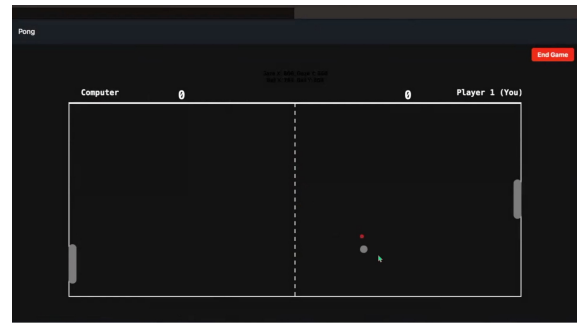
4.3 Dynamic Colour Shading

The cornerstone of this project is the implementation of dynamic colour shading as a method for passive colour vision testing. The colour schemes used are based on established research into colour blindness and were selected to reflect known confusion pairs for each condition.

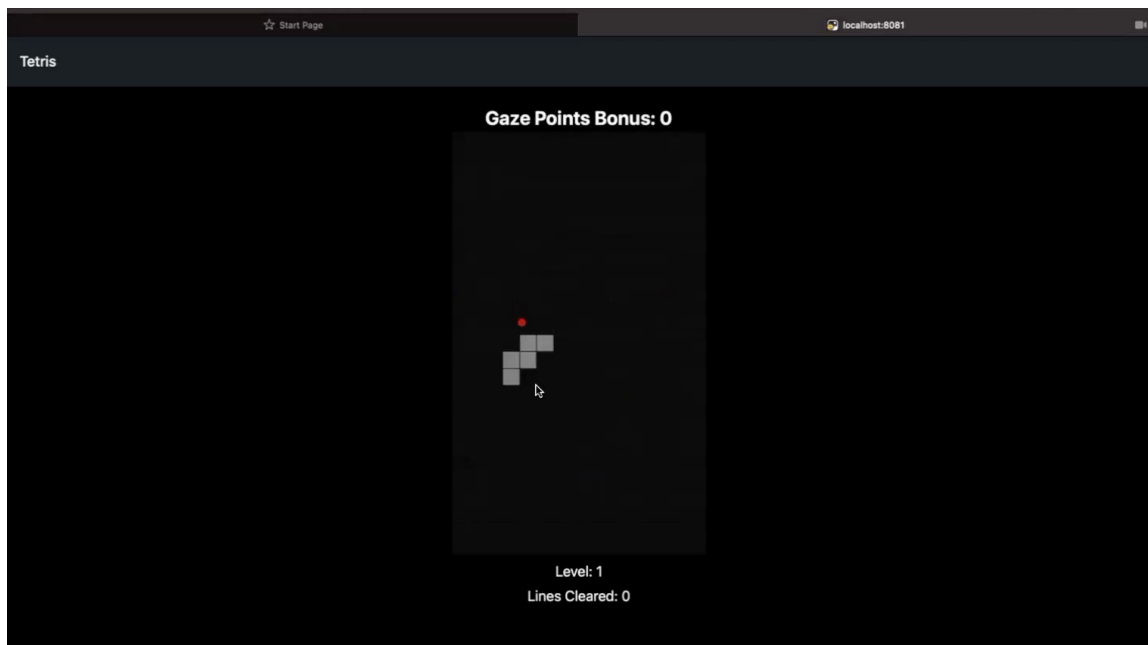
- **Monochromacy (achromatopsia):** Includes different shades of grey ranging from black to white.



(a) Space Invaders Greyscale Shade Pair



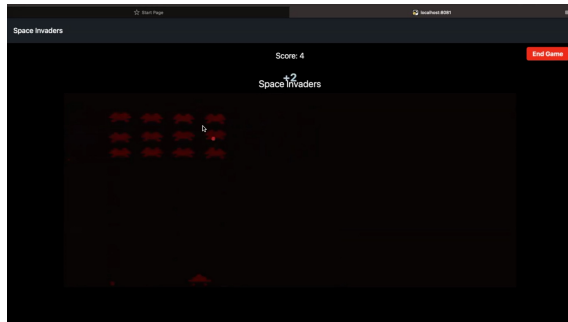
(b) Pong Greyscale Shade Pair



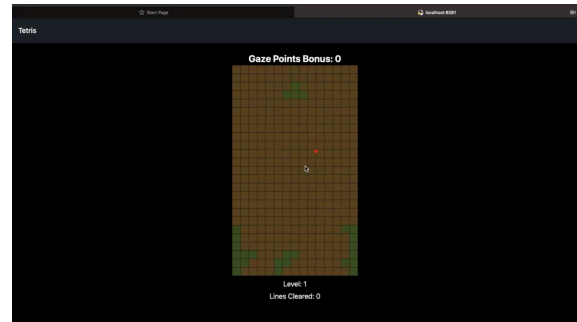
(c) Tetris Greyscale Shade Pair

Figure 4.5: Greyscale gameplay examples for Space Invaders, Pong, and Tetris under Monochromacy Colour Shading.

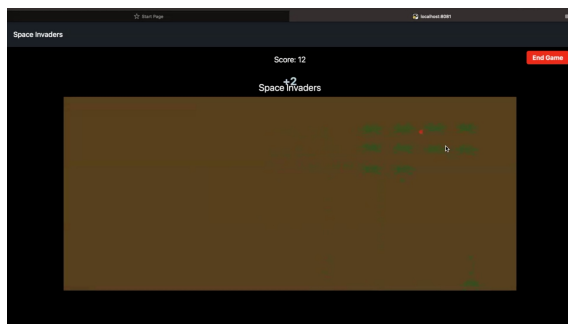
- **Protanopia (Red-Blindness):** Red elements are rendered in tones that become indistinguishable from green or brown.



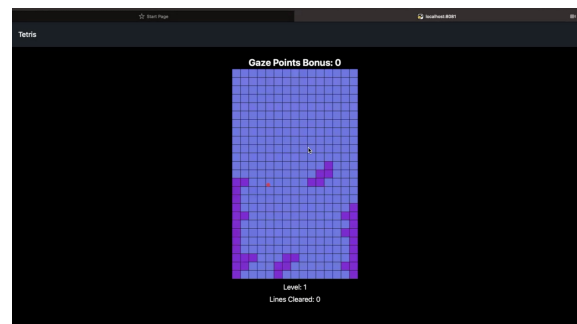
(a) Space Invaders Protanopia Shade Pair



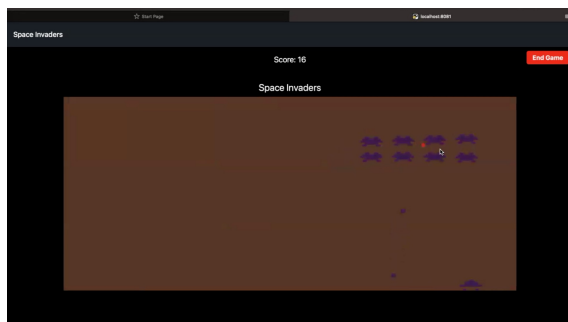
(b) Tetris Protanopia Shade Pair



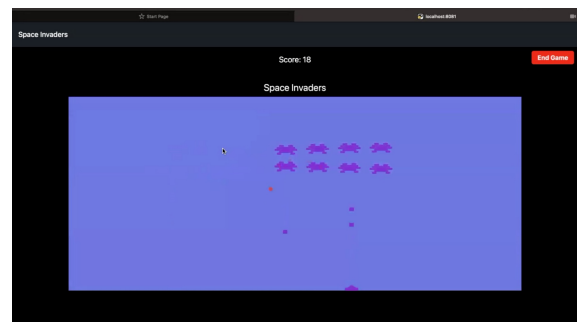
(c) Space Invaders Protanopia Shade Pair



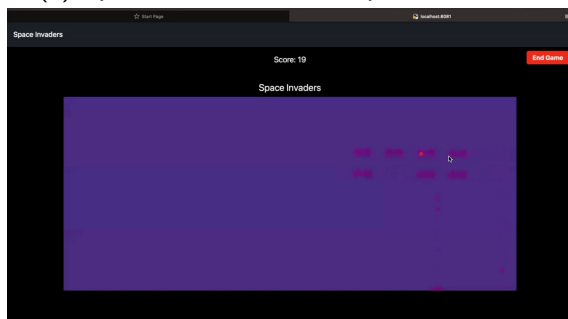
(d) Tetris Protanopia Shade Pair



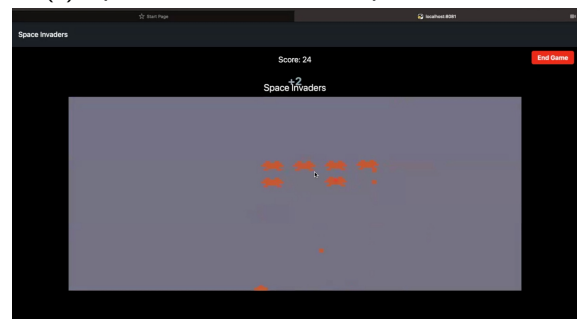
(e) Space Invaders Protanopia Shade Pair



(f) Space Invaders Protanopia Shade Pair



(g) Space Invaders Protanopia Shade Pair



(h) Space Invaders Protanopia Shade Pair

Figure 4.6: Protanopia Colour Mode Shade Pairs.

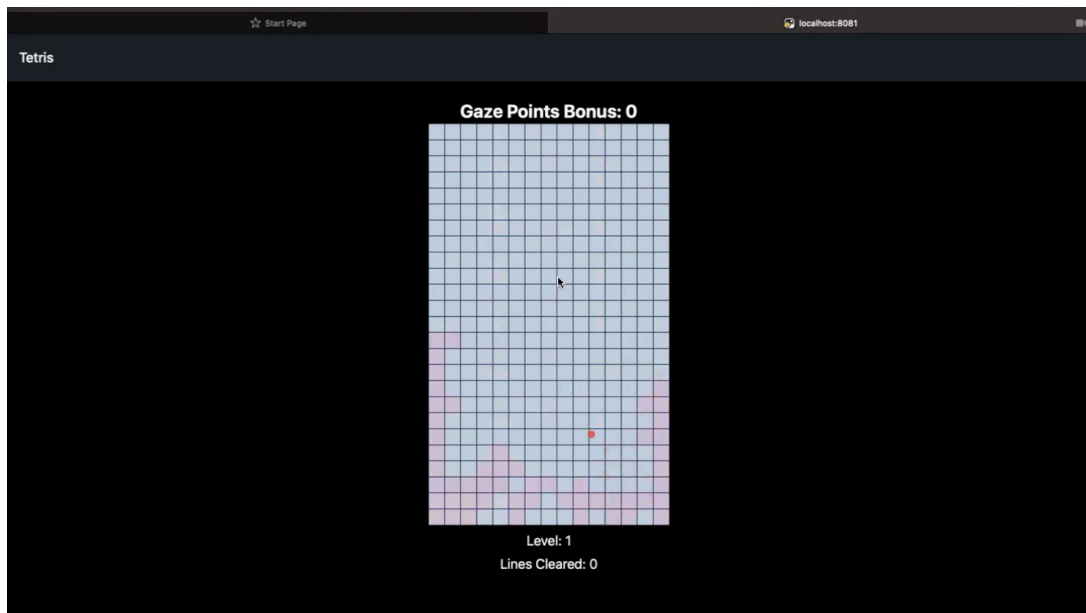
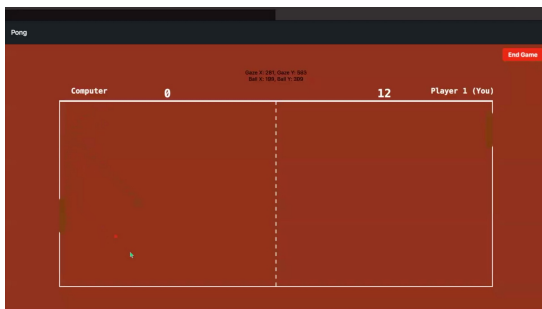


Figure 4.7: Tetris Protanopia Shade Pair.

- **Deuteranopia (Green-Blindness):** Colours are shifted to obscure the red-green contrast, replacing greens with hues problematic for those with medium-wavelength cone deficiency.



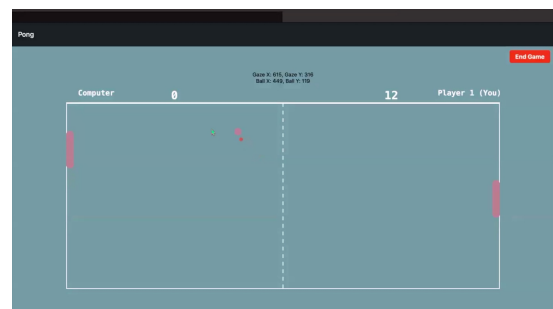
(a) Pong Deuteranopia Shade Pair



(b) Pong Deuteranopia Shade Pair

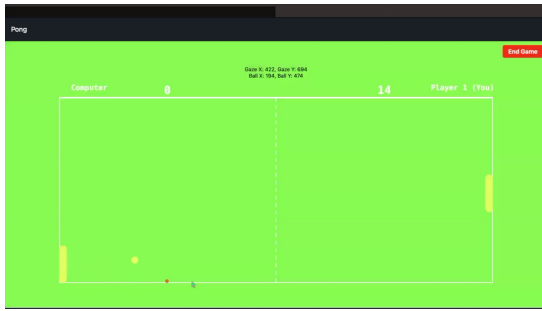


(c) Pong Deuteranopia Shade Pair



(d) Pong Deuteranopia Shade Pair

Figure 4.8: Pong Deuteranopia Shade Pairs.



(a) Pong Deuteranopia Shade Pair



(b) Pong Deuteranopia Shade Pair



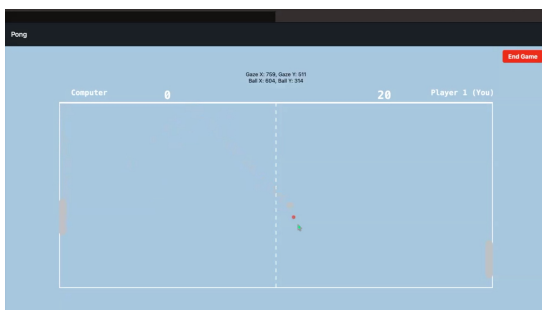
(c) Pong Deuteranopia Shade Pair



(d) Pong Deuteranopia Shade Pair

Figure 4.9: Pong Deuteranopia Shade Pairs (contd.).

- **Tritanopia (Blue-Blindness):** Blue and yellow hues are replaced with tones that converge into greys or greens, masking the typical blue-yellow distinction.



(a) Pong Tritanopia Shade Pair



(b) Pong Tritanopia Shade Pair



(c) Pong Tritanopia Shade Pair



(d) Pong Tritanopia Shade Pair

Figure 4.10: Pong Tritanopia shade pairs.

Each game initiates in a shade of the greyscale colour scheme. As gameplay progresses, the colours of each game entity, as well as the game background, are cycled through the predefined palettes corresponding to each CB type. This transition occurs gradually and without user notification to avoid influencing behaviour or drawing attention to the visual alteration.

During each game session, the application logs performance data, including score, bonus points, object interactions, timing data, and the colour mode active at the time. These metrics are exported for later analysis. A sustained decline in performance during a particular colour mode may suggest a correlation between the user's visual perception and a specific type of colour blindness.

This system offers a novel, engaging and non-intrusive method of screening users for potential colour vision issues while maintaining the integrity of gameplay and user experience.

4.3.1 Dynamic Colour Shading System in Code

The dynamic colour shading system in the games (Pong, Space Invaders and Tetris) is designed to gradually transition between curated colour pairs, each of which is chosen to either simulate or test for a specific type of colour blindness. The implementation uses a smooth interpolation technique to blend between colours in real-time, creating a continuous and subtle change that does not disrupt gameplay. This also serves a diagnostic purpose: by monitoring a player's performance or eye gaze during different colour modes, the system can infer potential colour vision deficiencies.

4.3.1.1 Colour Pair Definitions

The system begins with a JavaScript object named `colourBlindPairings`, which categorises colour pairs under four different visual modes according to the four colour blind types:

- `greyscale` – neutral black and white gradients for baseline calibration.
- `protanopia` – red-green deficiency.
- `deutanopia` – green deficiency.
- `tritanopia` – blue-yellow deficiency.

Each mode contains an array of primary and secondary RGB colours, as shown in Appendix A1.2.1.

4.3.1.2 Interpolating Colours

When switching from one colour pair to the next, there is a very subtle change in the game entities and background colours. This is done to ensure a gradual contrast between any two

sets of colour pairings. The function `interpolateTwoColours` handles interpolation between each colour pair, based on a time step counter, as seen in Appendix A1.2.2.

This function returns two colours:

- `backgroundColour` — weighted more towards the primary colour.
- `entityColour` — weighted more towards the secondary colour.

These are used to shade the game background and game entities (Pong ball and paddles).

4.3.1.3 Cycling Between Colours

The function `cycleColourPairWithTwoOutputs` calculates the ratio of transition and returns interpolated colours, as shown in Appendix A1.2.3.

Here, `ratio` normalises the current step to a value between 0 and 1. This ratio is then used to compute an interpolated colour value between the primary and secondary colours. This is done to ensure enough contrast between the two sets of colours in each pair (game entity and background) but also to withhold the integrity of each pair true to each CB type.

4.3.1.4 Applying Colours

In the main component, the current background and entity colour are computed using React's `useMemo` hook, as seen in Appendix A1.2.4.

This ensures that colour updates are only recalculated when the inputs change, improving performance and responsiveness.

4.3.1.5 Timing and Colour Mode Transitions

The dynamic shading is driven by an interval that ticks every 40 milliseconds, updating the step counter and rotating through colour pairs and modes accordingly, as seen at Appendix A1.2.5.

Each pair lasts for 100 steps (approximately 3 seconds), and the full cycle through all colour modes takes around three minutes. Once a mode completes, the system switches to the next mode in the cycle. This is done to give the user enough time to register the visual change in the game entity and to track the entity to the best of their ability.

4.3.1.6 Applying Colours to Game Entities

The dynamically computed colour values are then applied to game elements such as paddles, the ball, the background, Space Invader aliens and player, and Tetris blocks.

In the JSX-based implementation provided (commonly used in React Native with SVG rendering) as seen in Appendix A1.2.6, the `<Rect>` component draws a Pong paddle. The `fill` attribute is assigned the value of `formattedEntityColour`, which is an RGB string interpolated based on the current colour blindness mode and step in the transition cycle.

This integration of dynamically shaded colours into UI elements forms the core of the diagnostic gameplay design. It allows the game to both remain visually engaging and serve as a passive assessment tool for identifying colour vision deficiencies.

4.3.2 Benefits

This system has several advantages. It avoids abrupt colour changes, maintaining user immersion. The transitions are subtle enough to prevent user distraction, while still allowing the system to log reactions, performance, or gaze data under different colour conditions. This allows developers to gather evidence of potential colour vision deficiencies without the user being explicitly aware of the test.

4.3.3 Conclusion

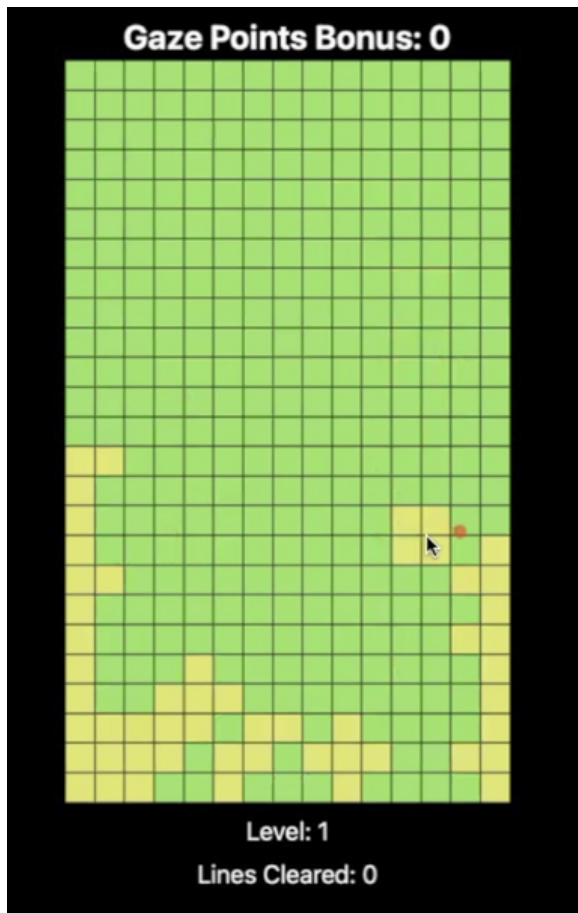
In summary, the dynamic colour shading technique offers an elegant, novel and non-invasive way of testing for colour blindness within the context of retro-style arcade games. Through careful timing, colour interpolation, and logical progression through different vision modes, this implementation creates both a playable and diagnostically useful experience for the user.

4.4 Colour Blind Game Simulation

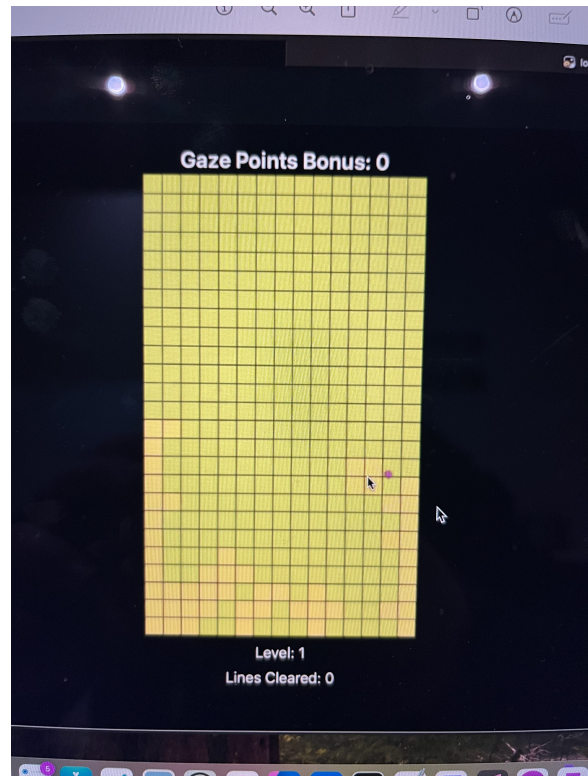
To illustrate the impact of colour blindness on gameplay, the Figures 4.11a and 4.11b show a comparison between normal vision and the simulated perspective of a player with Deuteranopia while playing Tetris.

Deuteranopia, a form of red-green colour blindness, makes it difficult to distinguish between colours that contain red or green components.

These figures highlight how certain Tetris blocks, which appear fairly distinct under normal vision, become nearly indistinguishable under the Deuteranopia simulation. This demonstrates the effectiveness of using games like Tetris for the gamification of colour blind screening.



(a) Normal Vision still from Tetris gameplay.



(b) Deuteranopia filter applied to same frame.

Figure 4.11: Comparison of Normal Vision and Deuteranopia macOS Simulation Filter in Tetris.

4.5 Eye-Tracking

Eye-tracking is incorporated into the web application. Its purpose is to add accuracy while measuring how precise the user's focus is in relation to certain game entities (e.g. Pong ball, Space Invader aliens) on the screen during gameplay as the colours change. If the user loses track of a game entity while a certain colour palette is in use, the eye-tracking will record the user's gaze data for later analysis.

Incorporating eye-tracking capabilities into the application was achieved through the integration of WebGazer.js, an open-source JavaScript library that enables real-time gaze estimation using standard webcams [55]. WebGazer.js operates entirely within the user's browser, ensuring that no video data is transmitted to external servers, thereby maintaining user privacy. The library requires user consent to access the webcam and employs a self-calibrating model that learns from user interactions, such as mouse clicks and cursor movements, to map eye features to screen positions. This approach eliminates the need for specialised hardware and allows for seamless embedding into web applications with minimal coding effort. By leveraging

WebGazer.js, the application can unobtrusively monitor users' gaze patterns during gameplay, facilitating the collection of data pertinent to colour differentiation and visual attention.

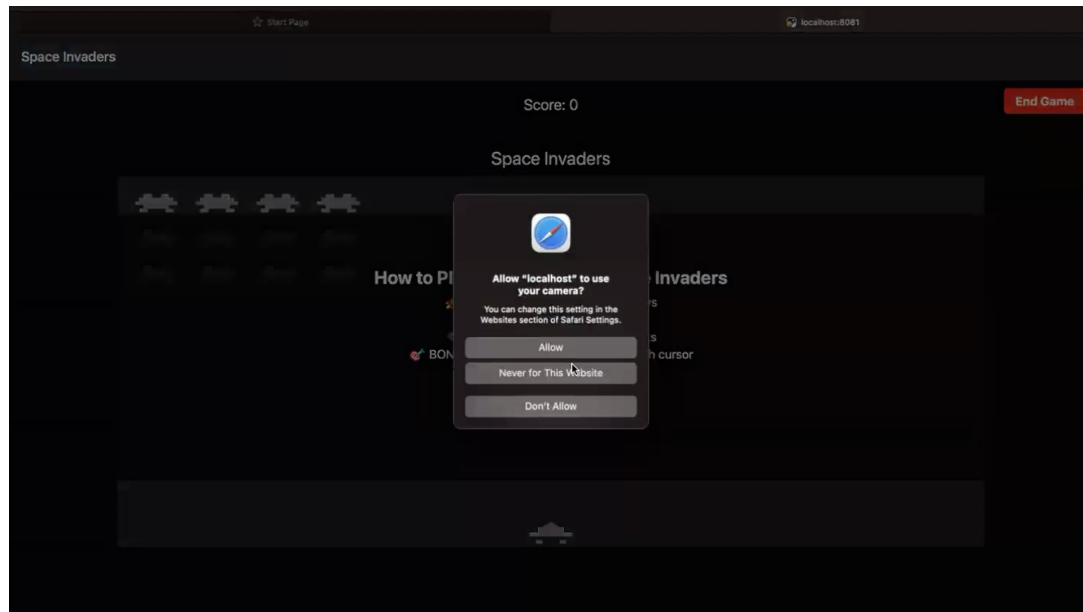


Figure 4.12: Pop-up for allowing Eye-tracking in the browser.



Figure 4.13: Eye-Tracking with red dot overlay for user's gaze in Pong.

4.6 Comparison of Eye-Tracking Technologies

In selecting a suitable eye-tracking technology for integration into this project, a comparative analysis was conducted on for prominent tools: GazeCloud API, WebGazer.js, MediaPipe and Tobii Pro SDK. Each of these solutions offers different trade-offs in terms of accuracy, accessibility, privacy, and implementation complexity.

4.6.1 GazeCloud API

GazeCloud is a commercial web-based eye-tracking solution that requires user permission to access the webcam and offers real-time gaze tracking via the browser [56]. It includes features such as heatmaps and visual attention metrics and is designed to work out-of-the-box with minimal configuration.

The primary advantage of GazeCloud is its plug-and-play nature and relatively high accuracy for a browser-based system. However, being a cloud-based service, it typically transmits gaze data to remote servers for processing. This raises privacy concerns, particularly for applications involving sensitive user data or requiring offline functionality. Additionally, usage is subject to rate limits and licensing constraints for full functionality, making it less desirable for open-source or academic projects. The free demo only gives access for a very limited time period, making it infeasible to use as a free tool.

4.6.2 WebGazer.js

WebGazer, introduced above, uses a self-calibrating machine learning model that maps facial feature movements to screen coordinates by learning from user interactions such as mouse clicks and cursor movement. While it does not achieve the same accuracy as commercial hardware solutions or deep-learning-based methods, its ease of integration and privacy-preserving architecture make it a highly suitable choice for lightweight applications, educational tools, and research experiments.

4.6.3 MediaPipe

MediaPipe, developed by Google, is a powerful open-source framework for building multimodal machine learning pipelines [57]. It includes a suite of models for hand, face, and body tracking. Although MediaPipe does not offer dedicated eye-tracking functionality out of the box, its FaceMesh module provides high-fidelity 3D facial landmarks, including those around the eyes.

By using the landmarks from FaceMesh, one can estimate gaze direction through vector geometry. This approach yields significantly higher accuracy compared to browser-based alternatives when run on modern devices with good camera input. However, MediaPipe is more computationally intensive and generally requires either mobile-native or desktop-native environments for optimal performance. Its integration into web applications is possible but requires WebAssembly (Wasm) or TensorFlow.js support, adding complexity to browser deployment.

4.6.4 Tobii Pro SDK

Tobii Pro SDK is a professional-grade software development kit used with Tobii's proprietary eye-tracking hardware [58]. It offers highly accurate and low-latency gaze data, with advanced features such as fixation mapping, saccade detection, and pupil dilation measurement. This makes it ideal for scientific research, usability testing, and clinical applications. However, it requires dedicated hardware, making it unsuitable for web or mobile apps that aim to reach a broad audience.

Feature	GazeCloud API [59]	WebGazer.js	MediaPipe	Tobii Pro SDK
Open Source	✗	✓	✓	✗
Client-Side Only	✗	✓	✓*	✗
Privacy Focus	✗	✓	✓	✓
Accuracy	Medium	Low–Medium	High	Very High
Ease of Setup	High	Medium	Low	Medium
Browser Support	✓	✓	✓**	✗
Real-Time Gaze	✓	✓	✓	✓
Hardware Required	✗	✗	✗	✓

Table 4.1: Comparison of Eye-tracking Technologies

* Requires WebAssembly or TensorFlow.js to run in the browser.

** MediaPipe browser use may require compilation or wrapper libraries.

Given the privacy and deployment constraints of this project, WebGazer.js was selected as the most appropriate solution. Its client-side architecture aligns well with the principles of user anonymity, and its lightweight footprint ensures compatibility and smooth game mechanics with React Native Web-based applications.

4.7 Design Decisions

These are noteworthy decisions that were made during the development stage of the project.

4.7.1 RGB Colour Space Vs. HLS Colour Space For Dynamic Colour Shading

The RGB colour space is a fundamental model used in digital imaging and visual display systems [60]. It is based on additive colour theory, where red, green, and blue light are combined in varying intensities to produce a broad spectrum of colours. Each channel (R, G, and B) typically ranges from 0 to 255 in 8-bit systems, enabling over 16 million unique colour combinations.

This model is widely applied in screens, monitors, digital cameras, and other visual technologies. However, the RGB colour space is device-dependent, meaning the same RGB values may yield different

visual results depending on hardware characteristics or calibration settings. To mitigate this, colour management systems are employed to ensure consistency and fidelity across different devices and display environments.

The RGB colour space is the standard for screens. Displays and GPUs work in this space, meaning that rendering in RGB is more efficient as it avoids unnecessary colour space conversions. Also, blending and shading are faster in RGB than in HLS because they work directly with linear intensity values, which is vital for my dynamic shading.

Conversely, the HLS colour space represents colours using three components: Hue, Lightness, and Saturation [61]. Hue denotes the type of colour and is measured in degrees from 0 to 360. In this model, 0° corresponds to blue, 60° to magenta, and 120° to red, with complementary colours situated 180° apart. Lightness indicates the brightness of the colour, ranging from 0.0 (black) to 1.0 (white). Saturation measures the purity of the hue; a saturation of 0 results in a shade of grey, while a saturation of 1 yields a fully saturated colour. The HLS model is visualised as a double hexcone, where any horizontal cross-section forms a hexagon. This colour space can be either device-dependent or device-independent, depending on its implementation.

The HLS colour space has discontinuities. The hue component in HLS is circular (ranging from 0° to 360°), which can cause discontinuities when interpolating colours. This has a negative affect on smooth colour transitions. Furthermore, linear blending of RGB colours is more predictable and stable, leading to smoother gradients and transitions in animations and shading. RGB is better for real-time lighting and shading. Realistic shading models (like Phong, Lambertian, or PBR) depend on linear RGB values for accurate light interaction. HLS does not directly correlate with physical lighting properties. HLS space can also lead to perceptual issues. Modifying lightness (L) in HLS does not always yield intuitive results, as it doesn't match human brightness perception accurately. Overall, RGB space provides more control over colour balance, which is essential for the dynamic colour shading element of this project.

4.8 Implementation Issues

4.8.1 Eye-Tracking Overhead

During the integration of WebGazer.js for real-time eye-tracking in the web application, several performance issues arose due to the high frequency of gaze data processing and the default configuration of the library. The original implementation of WebGazer samples gaze data at a relatively high frequency, typically up to 60 frames per second (fps), and makes use of webcam video streams at medium-to-high resolution. While effective for laboratory conditions, these defaults can lead to excessive CPU usage and memory consumption in standard browser environments, especially on lower-end devices or when multitasking.

To address this challenge—referred to here as eye tracking overhead—the implementation was refactored to reduce computational demands while maintaining acceptable tracking accuracy. Key al-

terations included throttling the gaze listener to process data at most once every 200 milliseconds (`THROTTLE_MS = 200`), rather than at the default rate. This significantly reduced the volume of gaze updates passed to the parent component, alleviating performance pressure.

Furthermore, the video input resolution was reduced to 160x120 pixels with a capped frame rate of 15fps, minimising the load on the browser's rendering and video decoding pipelines. Additionally, the prediction visualisation was simplified by disabling the face overlay and video preview, and the regression model was changed from `ridge` to `weightedRidge`, which provided a better balance between performance and stability.

A module-level flag (`webGazerInitialized`) was introduced to ensure that WebGazer is only initialised once per game session, avoiding repeated calls that previously led to memory leaks or unexpected behaviour. The full system was further protected using React's lifecycle hooks to ensure proper teardown using `webgazer.end()` on component unmount.

These modifications allowed the eye-tracking component to operate more efficiently across a range of devices and browsers, improving frame stability and reducing lag. However, it should be noted that these optimisations may slightly reduce the spatial accuracy of gaze predictions, a trade-off that was deemed acceptable in the context of the game's diagnostic rather than precision-tracking purpose.

4.8.2 Colour Settings Between Machines

A key implementation challenge encountered during development and early initial testing was the inconsistency in colour rendering across different operating systems, particularly between macOS and Windows. This issue stems primarily from the distinct default colour profiles and display calibration settings utilised by each system.

macOS devices typically use a Display P3 colour profile by default, which supports a wider gamut than the standard sRGB profile employed by most Windows machines [62]. As a result, colours that appear vibrant and well-contrasted on a Mac display may look noticeably muted or imbalanced on a Windows device. This discrepancy becomes especially problematic in my colour blind games, where precise colour transitions and contrast differences are critical to evaluating visual perception and potential colour blindness.

Furthermore, Mac displays often apply system-wide colour correction and gamma adjustments that enhance visual richness, whereas Windows relies more heavily on the monitor's own configuration [63]. These differences mean that the same RGB values may appear subtly or even significantly different depending on the platform, which undermines the consistency and reliability of the diagnostic environment intended by the dynamic shading system.

To mitigate this, I focused on using a macOS laptop to conduct initial testing for the study to keep results consistent across all participants. Testing was conducted on both platforms in the early development stages to evaluate how each colour mode performed under varied display conditions. However, a number of colour blind participants were observed playing the games on macOS in the

early stages and given accurate results, therefore I chose to use this system for the full testing set. Nonetheless, the implementation would benefit from further refinements such as allowing the user to calibrate their display during initial setup, or automatically detecting system colour profiles and adjusting the rendering pipeline accordingly.

4.9 Summary

This chapter outlined the technical design and implementation of the system developed to assess colour vision through interactive gameplay. It detailed the architectural structure of the application, the integration of eye-tracking technology, and the dynamic manipulation of colour palettes across multiple retro-style games. The rationale behind game selection, colour cycling techniques, and the incorporation of subtle testing mechanisms was discussed. By embedding diagnostic elements within familiar gameplay environments, the system aims to provide an engaging and non-intrusive means of identifying colour vision deficiencies. This foundation sets the stage for evaluating user interaction and system performance, which is explored in the following chapter on user testing.

5 User Testing

5.1 Participation Recruitment

Participants were recruited for this project through an online Google Forms anonymous survey. The participant pool included students from the School of Computer Science and Statistics. This survey included the Participation Information Leaflet for potential participants with details on the study including its purpose and conditions for involvement.

Both non-colour blind and colour blind participants were invited to enrol in this study. The non-colour blind participants act as the control for my study. Non-colour blind people help define how a person without a colour vision deficiency should perform on the test (e.g. accuracy, reaction time, mistake patterns, gaze tracking), giving a reference point to spot deviations. If non-colour blind users also struggle with the same colour combinations, then the test is not specific enough. This helps to detect false positives (people incorrectly flagged as colour blind).

5.2 Participation Details

This study involved 50 participants who all played each colour blind game. The anonymous survey indicated that 10 of these had been previously diagnosed with colour blindness by a medical professional.

Participants were allowed to give verbal feedback during testing to capture their real-time thoughts, perceptions and any moments of confusion or difficulty they had with identifying game entities. This think-aloud approach provided valuable qualitative insights that complemented the quantitative gameplay data. It enabled the identification of specific colour pairings that caused hesitation or misidentification.

6 Results and Evaluation

6.1 Evaluation Approach

6.1.1 Eye-Tracking Scoring System

The user plays the games as the colours change. The Webgazer.js eye-tracking works by self-calibrating, in real-time, according to the mouse pointer position on screen. As the user moves the mouse pointer and watches the game entity simultaneously, Webgazer calibrates the eye-tracking according to these movements, essentially supplemented by the user's interaction with the mouse pointer.

To compensate for the relatively poor performance of the eye-tracking implementation I used, which could be further improved by using a dedicated eye-tracking hardware peripheral or a premium software implementation of eye-tracking technology - see Section 4.6.4, I instructed players to look at and track the game entities (Pong ball, Space Invaders aliens, Tetris block) as they move on the screen with the mouse pointer in order to provide the most accurate eye-tracking data. If the user takes time to find the game entity or they stop moving the cursor because they can't identify it, this is recorded.

The player is rewarded with bonus points when they track the various game entities closely over a period of time. As shown in Figure 6.1, the user is rewarded two bonus points with an animation during a greyscale shade pair after closely tracking the alien entities area on screen for a period of time.

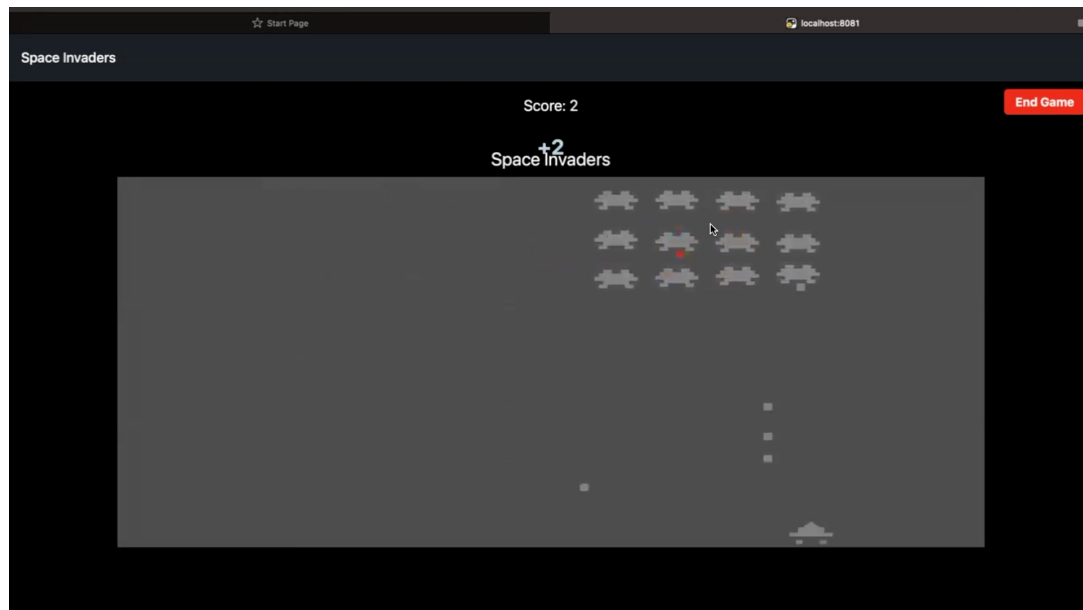


Figure 6.1: Eye-tracking System Awarding Two Bonus Points to User during Space Invaders Gameplay.

6.1.2 Results CSV File

I export a Comma Separated Value (CSV) file when the user clicks on the 'End Game' button once the test is over for each game (shown in Figure 6.1 above). This file contains data recorded by the eye-tracking system as well as the number of bonus points awarded to the player over the course of each game for closely tracking the different game entities. Specifically, it captures:

- Participant Number
- The diagnosis for each participant per game from this study
- Recorded Eye-tracking Accuracy over the course of each game per participant
- The count of Bonus Points earned per participant per game for close gaze tracking (maximum of 155 available per game)
- Percentage of Bonus Points Won by Participant against Bonus Points Available

6.2 Evaluation

6.2.1 Game-Specific Results

The table below shows the full results over all three games for the 50 study participants.

Participant	Pong	Space Inv.	Tetris	Eye-tracking Acc.%	Count Bonus Pt.	Bonus Pt.%
1	No CB	No CB	No CB	100%	155	100%
2	No CB	No CB	No CB	100%	145	93.55%
3	No CB	No CB	No CB	100%	146	94.19%
4	No CB	No CB	No CB	100%	148	95.48%
5	No CB	No CB	No CB	100%	152	98.06%
6	No CB	No CB	No CB	100%	153	98.71%
7	No CB	No CB	No CB	100%	150	96.77%
8	No CB	No CB	No CB	100%	149	96.13%
9	No CB	No CB	No CB	100%	148	95.48%
10	No CB	No CB	No CB	100%	146	94.19%
11	No CB	No CB	No CB	100%	147	94.84%
12	No CB	No CB	No CB	100%	152	98.06%
13	No CB	No CB	No CB	100%	153	98.71%
14	No CB	No CB	No CB	100%	155	100%
15	No CB	No CB	No CB	100%	152	98.06%
16	No CB	No CB	No CB	100%	149	96.13%
17	D+P	D+P	D+P	50%	47	30.32%
18	No CB	No CB	No CB	100%	150	96.77%
19	No CB	No CB	No CB	100%	149	96.13%
20	No CB	No CB	No CB	100%	148	95.48%
21	No CB	No CB	No CB	100%	146	94.19%
22	No CB	No CB	No CB	100%	147	94.84%
23	No CB	No CB	No CB	100%	152	98.06%
24	No CB	No CB	No CB	100%	153	98.71%
25	No CB	No CB	No CB	100%	149	96.13%
26	No CB	No CB	No CB	100%	147	94.84%
27	D	D	D	75%	95%	61.29%
28	No CB	No CB	No CB	100%	152	98.06%
29	No CB	No CB	No CB	100%	155	100%
30	D+P	D+P	D+P	50%	45	29.03%
31	D	D	D	75%	92	59.35%
32	D	D	D	75%	96	61.94%
33	No CB	No CB	No CB	100%	148	95.48%
34	D	D	D	75%	94	60.65%
35	D	D	D	75%	93	60%
36	No CB	No CB	No CB	100%	148	95.48%
37	No CB	No CB	No CB	100%	146	94.19%
38	No CB	No CB	No CB	100%	147	94.84%
39	No CB	No CB	No CB	100%	152	98.06%

Participant	Pong	Space Inv.	Tetris	Eye-tracking Acc.	Count Bonus Pt.	Bonus Pt. %
40	No CB	No CB	No CB	100%	153	98.71%
41	D	D	D	75%	96	61.94%
42	No CB	No CB	No CB	100%	147	94.84%
43	D	D	D	75%	94	60.65%
44	D	D	D	75%	93	60%
45	No CB	No CB	No CB	100%	148	95.48%
46	No CB	No CB	No CB	100%	146	94.19%
47	No CB	No CB	No CB	100%	147	94.84%
48	No CB	No CB	No CB	100%	152	98.06%
49	No CB	No CB	No CB	100%	153	98.71%
50	No CB	No CB	No CB	100%	149	96.13%

Legend: No CB = No Colour Blindness, D = Deuteranopia, P = Protanopia, D+P = Combination of Deuteranopia and Protanopia, **Eye-tracking Acc.%** = Percentage of time gaze was within target zones during tracking period, **Count Bonus Pt.** = Gaze-based bonus points earned, **Bonus Pt.%** = Bonus points as % of max possible.

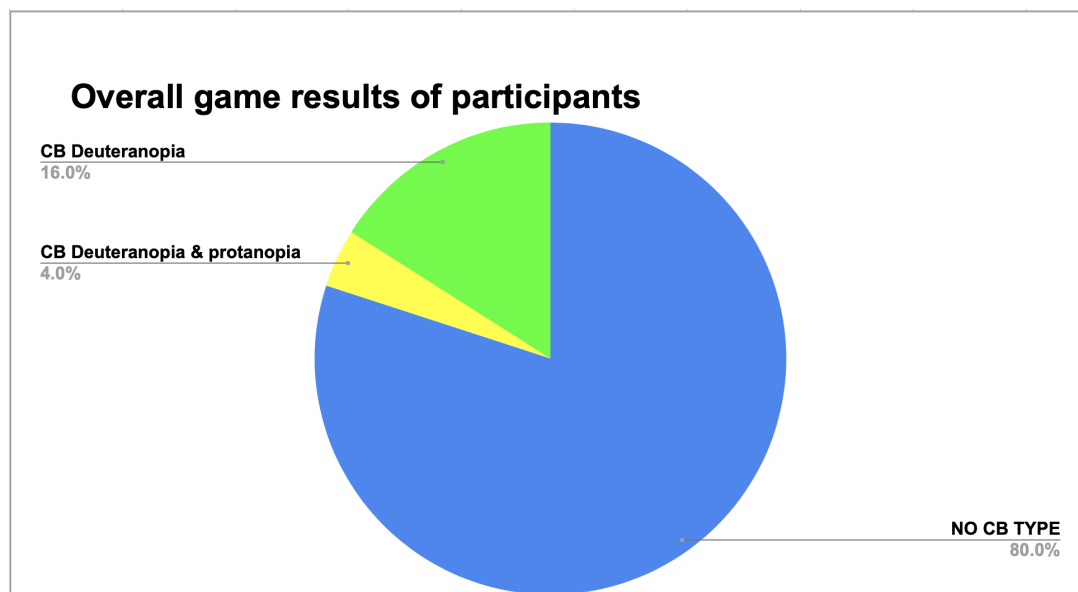


Figure 6.2: Pie Chart Showing the Different Proportions of Screening Results from Total Study Population.

From the pie chart above, it can be clearly seen that 40 of 50 non-CB study participants were accurately identified as having no CB type (80%). 8 of the 10 CB identified cases were found to be of the Deuteranopia type (16% of total study population) and 2 of the 10 CB identified cases were found to have a combination of both Deuteranopia and Protanopia CB types (4% of total study population).

Bonus Points earned as a % of total by participant

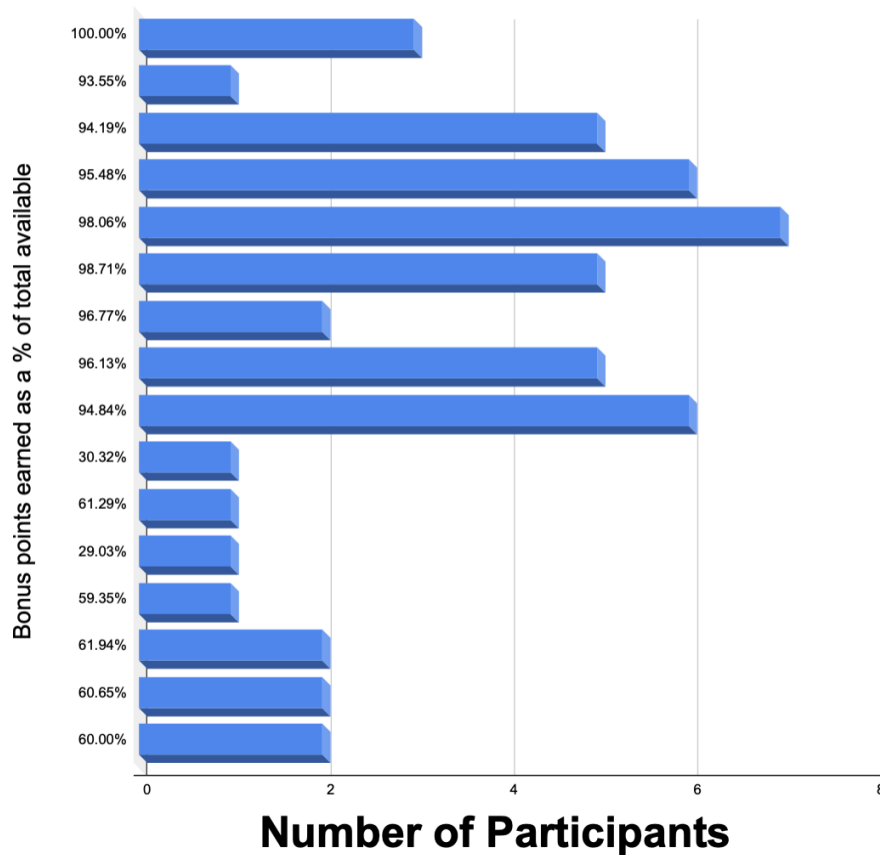


Figure 6.3: Bar Chart Showing the Bonus Points Earned as Percentage of Total by Participant.

From Figure 6.3,

- 3 participants had 100% success in bonus points
- 2 participants had approx. 30% success - these were the individuals with two types of CB
- 8 participants had approx. 60% success in bonus points - these were the individuals with Deuteranopia
- All 40 non-CB participants had a success rate of 95-100% in achieving max bonus points of 155

6.2.1.1 Colour Blind Diagnosis

Participant	Diagnosis Type from This Study	Previously Diagnosed Type
17	Deutanopia	Deutanopia
27	Deutanopia	Deutanopia
30	Deutanopia	Deutanopia
31	Deutanopia	Deutanopia
32	Deutanopia	Deutanopia
34	Deutanopia and Protanopia	Deutanopia
35	Deutanopia and Protanopia	Deutanopia
41	Deutanopia	Deutanopia
43	Deutanopia	Deutanopia
44	Deutanopia	Deutanopia

Table 6.2: Pong Screening Results

Participant	Diagnosis Type from This Study	Previously Diagnosed Type
17	Deutanopia	Deutanopia
27	Deutanopia	Deutanopia
30	Deutanopia	Deutanopia
31	Deutanopia	Deutanopia
32	Deutanopia	Deutanopia
34	Deutanopia and Protanopia	Deutanopia
35	Deutanopia and Protanopia	Deutanopia
41	Deutanopia	Deutanopia
43	Deutanopia	Deutanopia
44	Deutanopia	Deutanopia

Table 6.3: Space Invaders Screening Results

Participant	Diagnosis Type from This Study	Previously Diagnosed Type
17	Deutanopia	Deutanopia
27	Deutanopia	Deutanopia
30	Deutanopia	Deutanopia
31	Deutanopia	Deutanopia
32	Deutanopia	Deutanopia
34	Deutanopia and Protanopia	Deutanopia
35	Deutanopia and Protanopia	Deutanopia
41	Deutanopia	Deutanopia
43	Deutanopia	Deutanopia
44	Deutanopia	Deutanopia

Table 6.4: Tetris Screening Results

The first CB test participant (number 17) had been previously diagnosed with the deutanopia CB type. As they played the games, it became obvious through their game performance and verbal feedback that they were having trouble with identifying (tracking) the desired game entities for the first colour pairing for the deutanopia type (mid-red to mid-brown shade). As they continued playing the game through the other types, their performance notably increased, easily picking up bonus points through all tritanopia colour shades and again through greyscale pairs as the colour shading continued cycling. Once the same pair of mid-red to mid-brown appeared again, they repeated that they were having severe difficulty in locating the Pong ball, Space Invaders aliens and Tetris blocks on the screen.

Similar scenarios were noted for CB participants 27, 30, 31 and 32. They were all able to easily identify the game entities throughout the greyscale and protanopia colour pairings, although I noted that from the protanopia dark-brown to dark-red and mid-green to muted-orange pairings, participant 30 and 31 required to lean forward slightly to keep tracking. Nonetheless they stated that they could still track the game entities during these pairs, which was reflected in their results. However, they completely lost track of the game entities for the first shade pairings of deutanopia on both colour cycles. They had no problems tracking the game entities for the tritanopia shade pairings. These were all reported to have moderate deutanopia colour blindness.

CB Participant 34 had no problems with tracking game entities for all greyscale shade pairings. However, they began to lose track of game entities during some protanopia shades. They verbally confirmed that they had trouble identifying for the dark-brown to dark-green, cornflower-blue to dark-violet and mid-green to muted-orange shade pairings. This participant also could not identify the game entities for the deutanopia shade pairing of mid-red to mid-brown shade, the shade pairing of blue-green to mid-pink and the pairing of pale-pink to light-grey. The results for this participant showed deficiencies in deutanopia and protanopia types, more severe of deutanopia than protanopia, after a previous diagnosis of deutanopia only. This is possible to occur. A person can have a combination of red-green and blue-yellow deficiencies, leading to a more complex pattern

of colour perception. For example, someone might have both protanopia and tritanopia, which would significantly impact their ability to perceive colours. This shows that my screening tool is very effective at identifying mild and severe cases of colour blindness between the CB types.

The same situation was observed of CB participant 35. They could see all shades of greyscale and tritanopia. However, they struggled to track game entities for the same shades of protanopia and deuteranopia as participant 34. Verbal feedback from this participant confirmed this, as they stared blankly at the screen for these select shades and stated that they thought the game had glitched or broken until they could see the game entities again for the remaining shades of colour modes. This participant had also only been diagnosed before with deuteranopia, and received a diagnosis of more severe deuteranopia to mild protanopia, further attributing to my screening tool's accuracy.

CB Participant 41 mentioned before the test that they had been previously diagnosed with colour blindness, but was not exactly sure what type. They thought it was deuteranopia from the Ishihara plates test they had taken years prior, but stated they were not certain. During the games, they appeared to only struggle with tracking game entities for the first shade pair of deuteranopia (mid-red to mid-brown), which had become a common pattern among participants that my study revealed to be of moderate deuteranopia. Their screening results confirmed moderate deuteranopia for their colour blindness. To further validate this, after giving the results, the participant contacted a family member, who confirmed the actual type of deuteranopia from this previous Ishihara plates test.

The final two CB participants, number 43 and 44, also showed signs of moderate deuteranopia during gameplay. They struggled to track game entities during the same deuteranopia pair of mid-red to mid-brown. Post screening, they both took online Ishihara plate tests during their study time slots. Both Ishihara tests returned moderate deuteranopia colour blindness, further amplifying my screening tool's accuracy.

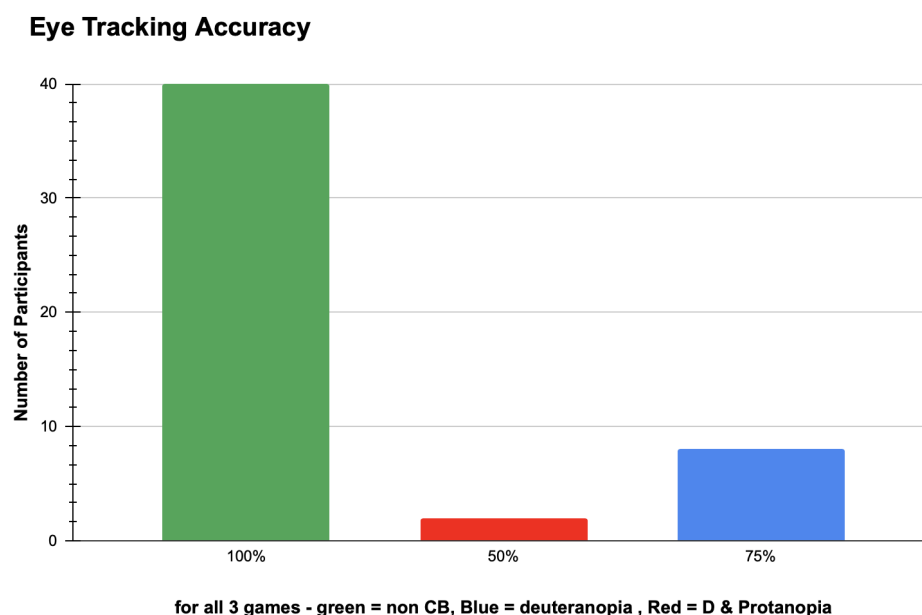


Figure 6.4: Bar Chart showing Eye-tracking Accuracy across Study Participants.

Figure 6.4 shows the eye-tracking accuracy for study participants across all three games by CB diagnosis.

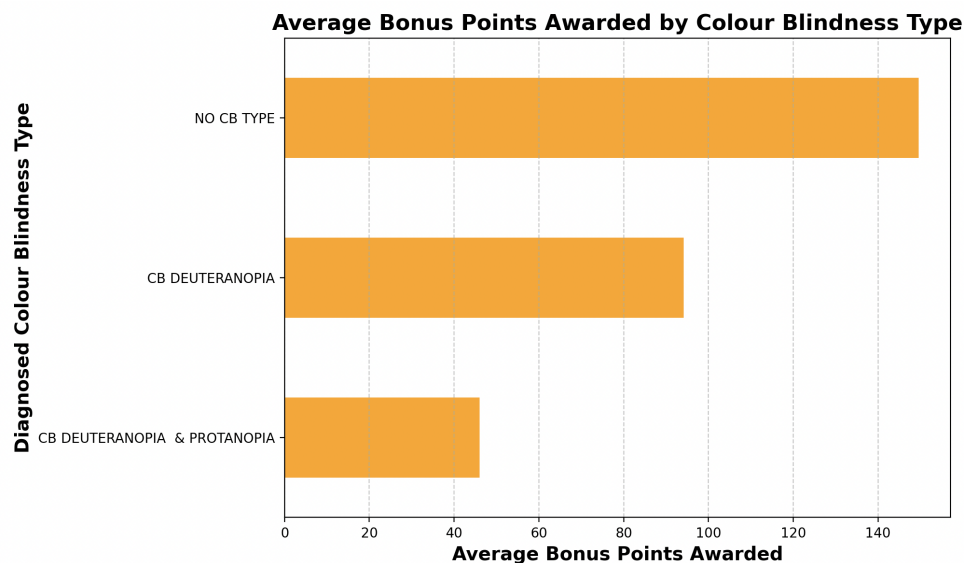


Figure 6.5: Bar Chart Showing Average Points Awarded by CB Diagnosis.

Figure 6.5 shows the average amount of bonus points earned by study participants across all three games by CB diagnosis.

6.2.1.2 Non-Colour Blind Diagnosis

All 40 of the participants identified as showing no signs of colour blindness were able to easily identify and track game entities throughout each game and colour cycle. Of those who offered verbal feedback, none of them reported difficulty tracking game entities.

As expected, this group was perfect to act as a control for the study.

6.2.2 Overall Results from Testing

Test results reveal that the colour blind games accurately identified 10 cases of colour blindness from all 50 study participants, with these cases all having previously been diagnosed with the condition (100% accuracy in diagnosing colour blindness overall).

In all of these 10 cases of colour blindness, the screening accurately identified the colour blind type (100% accuracy at identifying CB type).

One common pattern that emerged from testing was the diagnosis of deuteranopia CB. This was expected considering that the most common form of CB is red-green colour blindness, which includes deuteranopia.

6.2.3 Verbal Feedback from Non Colour Blind Participants

Non colour blind participants commented on the ease of play of the games, saying they were very enjoyable and easy to pick up. They were impressed by the logic behind subtly changing game colours to identify CB types. Some of these participants did suggest making some colour transitions less sharp on the eyes for brighter shades like yellow, but overall they were pleased with the presentation of the games.

One non CB participant was present for a second test trial post testing of a CB participant, with the permission of the CB participant. The non CB participant was amazed at the CB participant's inability to identify and distinguish between certain shades of deuteranopia, which the non CB participant found easy to identify through verbal feedback.

6.2.4 Verbal Feedback from Colour Blind Participants

One common comment from CB participants was their surprise regarding the accuracy of the results given the simplicity of the games. Many of them were shocked when they were unable to identify game entities on screen as they know the retro games used in their standard format and would have no issues playing them beforehand. They enjoyed playing the games and never stated that they felt they were being tested. Due to the smooth and quick transition of the games through the colour shading cycles, none of the participants were left too long unable to effectively play the game, as each shade pair only lasted three seconds. The majority of CB participants were unaware how quickly the screening had occurred. I had to inform many of them that the game had completed several cycles of colour pairs and that the test had completed without them noticing. Many of them were taken aback by its ease of use.

With written permission, I was allowed to record a CB participant playing the games. Audio and device display was recorded, without webcam. They provided commentary during gameplay of which colour pair shades were easy and difficult for them to see, and more importantly when they couldn't track the desired game entity at all. This was very valuable verbal feedback to support the accuracy of my screening tool in identifying colour blindness.

6.3 Summary

This chapter presented a detailed analysis of the screening tool's performance based on user interaction and testing data. The results demonstrated that the screening tool achieved a 100% success rate in identifying whether a user exhibited signs of colour vision deficiency, with a 100% success rate in correctly determining the specific type. These findings highlight the diagnostic potential of embedding adaptive visual challenges within retro-style gameplay. The evaluation also considered user engagement, system usability and the influence of eye gaze tracking on performance metrics. Limitations and unexpected patterns were discussed, offering valuable insights into how the tool can be refined further. These outcomes inform the concluding reflections and suggestions for future research presented in the final chapter.

7 Conclusion

This chapter summarises the work undertaken as part of this dissertation. It also discusses the primary weaknesses and recommended improvements together with recommended future works and the final outline from this work.

7.1 Objective Assessment

The primary research aim of this dissertation was to identify colour blindness using computer vision with video games. Ultimately, the testing showed empirical evidence that this novel method of hidden colour blind testing in video games is an effective means of diagnosis. The system has a 100% accuracy rate at identifying colour blindness, with a 100% success rate at identifying colour blind type.

7.2 Criticisms and Limitations

While the screening tool performed extremely well, there are several limitations that hindered its overall ability to determine CB.

Firstly, the eye-tracking functionality could be further developed to make gameplay easier. The refactoring of the Webgazer.js API reduced its gaze tracking performance considerably. My initial concern that the constant self-calibration of the eye-tracking with the user's mouse pointer would prove problematic. However, as the results showed, the users were clearly not affected by this.

Secondly, due to the overhead of the Webgazer.js original compute power for eye-tracking, using the web application within Google Chrome and Firefox was problematic, slowing or breaking gameplay mechanics altogether. However, everything functioned as intended running the application in Apple Safari.

7.3 Future Work

While this project demonstrates the feasibility of using video games and computer vision to screen for colour blindness, there are several avenues for future research and development to enhance the tool's accuracy, accessibility, and applicability.

7.3.1 Larger-Scale Testing with Diverse User Groups

The current testing was limited to a small number of adult participants, primarily from a university environment. To ensure broader reliability and accuracy, future iterations of the tool should be validated on a more diverse population, including a wider range of ages (especially a younger audience), ethnic backgrounds and confirmed colour blindness diagnoses. This would also include users with less common types of colour blindness, such as Protanopia and Tritanopia, to confirm the tool's effectiveness across the spectrum of visual impairments.

7.3.2 Game Diversification for Broader Visual Contexts

Expanding the types of games used in the screening process could offer a more robust testing environment. Different game genres place varying demands on colour perception and could help identify visual difficulties more effectively. For example, platformers, puzzle games, and action games might require users to interpret colours in motion or under time pressure. Introducing game variety can also keep the experience engaging, especially for younger users, and make the testing process feel more natural and enjoyable.

7.3.3 Improved Dynamic Colour Calibration

The current system uses a pre-defined set of colour pairs to trigger potential confusion in colour blind users. However, these colour pairs may not be optimally tuned to every individual's perception. Future versions of the tool could implement a more refined dynamic colour calibration mechanism that adjusts the shades and contrasts of colour pairs in real time based on user performance. This would help target specific weaknesses with greater precision and potentially allow the system to differentiate between mild and severe forms of colour blindness.

7.3.4 Integration of Eye Tracking and Reaction Timing

Incorporating more detailed behavioural data, such as eye gaze tracking and reaction time, could provide additional insights into user uncertainty or hesitation — even when the correct response is ultimately given. Subtle metrics such as delayed decision-making or repeated shifts in visual attention could be early indicators of visual confusion, offering a more nuanced approach to detecting borderline or latent forms of colour blindness.

7.3.5 Mobile App Deployment for Accessibility

To broaden the tool's reach and practical usability, future development should focus on creating a mobile-friendly version of the system. A mobile app would allow for deployment in primary schools, clinics, or even in home environments, particularly in under-resourced or rural areas. Including offline functionality and ensuring data privacy would be essential for ensuring accessibility while maintaining user trust.

7.3.6 Comparison with Traditional Screening Tools

Within the study, I was able to directly compare my results with the diagnosis from an online Ishihara Plates test. I was pleased that the results were mirrored. Nonetheless, to scientifically validate the tool's accuracy and diagnostic value, it would be beneficial to compare its results against more established colour blindness screening methods such as the Farnsworth D-15 test, or Cambridge Colour Test. By assessing metrics such as sensitivity, specificity, and false-positive rates, the game-based tool could be benchmarked against clinical standards, paving the way for potential use in medical settings.

7.3.7 Machine Learning for Personalised Classification

As more gameplay data are collected, machine learning algorithms could be employed to identify patterns in performance and predict a user's type of colour blindness. These models could take into account colour errors, gaze behaviour, and reaction times to classify users more accurately than rule-based systems. Over time, the tool could build a colour perception profile for each individual, which could then be used to personalise both testing and user interfaces.

7.3.8 Collaboration with Educational and Medical Institutions

Partnering with schools, optometrists, and healthcare providers could help bring this tool into real-world screening contexts. Such collaborations would allow for user testing in a more structured setting and could provide valuable feedback from educators and clinicians. Additionally, seeking ethical approval and considering medical certification standards could position the tool for use in official screening programmes.

7.3.9 Personalised Accessibility Tools Post-Screening

One promising extension of this tool is to not only diagnose but also assist. Once a user's colour vision profile is identified, the system could automatically apply colour adjustments in educational software, games, or web environments to improve accessibility. Customised visual settings — such as colour remapping, pattern overlays, or audio cues — could help users better interpret information in colour-dependent interfaces, fostering greater digital inclusion.

These future directions offer a pathway to evolve the prototype into a clinically valid, widely accessible, and user-friendly tool. They also highlight the interdisciplinary potential of combining game design, computer vision, and accessibility engineering to support inclusive digital environments.

7.4 Final Thoughts

Due to its ease of use, accessibility, accuracy and element of fun, this colour blind screening tool would be an ideal option for use by educators in early childhood settings, healthcare workers and optometrists. The fact that the participants do not feel any pressure while playing the games ensures greater accuracy of the results. The earlier the condition of colour blindness is identified, the more beneficial it is for the child, their family and their educators.

There is also a great opportunity for its use in the field away from a clinical setting, e.g. on a mobile device in developing countries. Naturally, there is a lot of work that can be done to expand on the work shown in this dissertation.

From its initial results (100% success rate in diagnosis), this project has shown signs of promise given its short time frame for development. Clinical tests typically progress over the span of several years [64], with many phases of testing and reiterations of development. In the span of around 7 months, this dissertation brought about an easy to use, engaging and accurate screening tool for colour blindness that can be used with anyone, especially children. The element of hidden testing within retro video games eliminates the feeling of being tested, ensuring that users receive the most precise results possible.

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A1 Appendix

A1.1 Github Code

The code for this dissertation can be found at the following url:

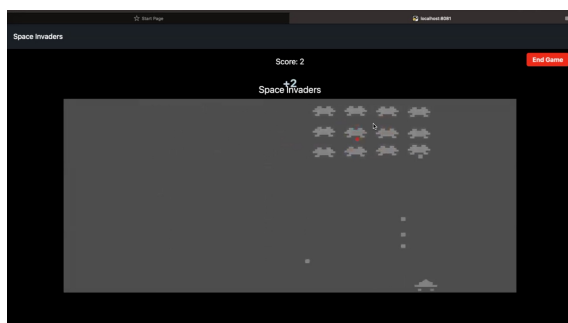
<https://github.com/JamesOC3310/thesis-project>

It is a private repository.

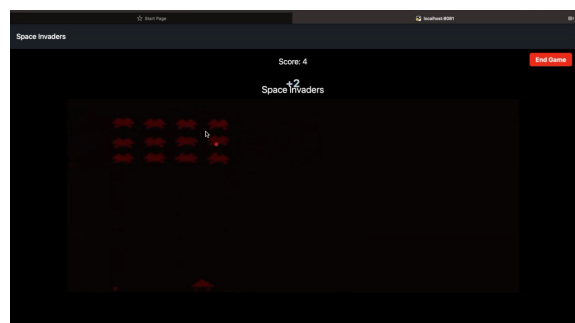
A1.2 Dynamic Colour Shading Code

A1.2.1 Colour Pairings

```
const colourBlindPairings = {  
  greyscale: [  
    { primary: { r: 30, g: 30, b: 30 }, secondary: { r: 220, g: 220, b: 220 } },  
    ...  
  ],  
  protanopia: [  
    { primary: { r: 15, g: 15, b: 15 }, secondary: { r: 220, g: 20, b: 60 } },  
    ...  
  ],  
}
```



(a) Space Invaders Greyscale Shade RGB Pair from Colour Pairings Code Snippet



(b) Space Invaders Protanopia Shade RGB Pair from Colour Pairings Code Snippet

A1.2.2 Interpolating Colours

```
function interpolateTwoColours(colour1, colour2, ratio) {
  const intermediateColour1 = {
    r: Math.round(colour1.r + (ratio / 2) * (colour2.r - colour1.r)),
    g: Math.round(colour1.g + (ratio / 2) * (colour2.g - colour1.g)),
    b: Math.round(colour1.b + (ratio / 2) * (colour2.b - colour1.b)),
  };

  const intermediateColour2 = {
    r: Math.round(colour1.r + (0.5 + ratio / 2) * (colour2.r - colour1.r)),
    g: Math.round(colour1.g + (0.5 + ratio / 2) * (colour2.g - colour1.g)),
    b: Math.round(colour1.b + (0.5 + ratio / 2) * (colour2.b - colour1.b)),
  };

  return { backgroundColour: intermediateColour1, entityColour: intermediateColour2 };
}
```

A1.2.3 Cycling Between Colours

```
function cycleColourPairWithTwoOutputs(pair, step, maxSteps) {
  const ratio = (step % maxSteps) / maxSteps;
  return interpolateTwoColours(pair.primary, pair.secondary, ratio);
}
```

A1.2.4 Applying Colours

```
const { backgroundColour, entityColour } = useMemo(() => {
  return cycleColourPairWithTwoOutputs(
    colourBlindPairings[colourMode][currentPairIndex],
    step,
    maxSteps
  );
}, [colourMode, currentPairIndex, step]);
```

A1.2.5 Timing and Colour Mode Transitions

```
...
const interval = setInterval(() => {
  setStep((prevStep) => {
    const newStep = prevStep + 1;

    if (newStep % stepsPerPair === 0) {
```

```

const pairs = colourBlindPairings[colourMode];
setCurrentPairIndex((prevIndex) => {
  if (prevIndex + 1 < pairs.length) return prevIndex + 1;

  setColourMode((prevMode) => {
    if (prevMode === "greyscale") return "protanopia";
    if (prevMode === "protanopia") return "deutanopia";
    if (prevMode === "deutanopia") return "tritanopia";
    return "greyscale";
  });

  return 0;
});

return newStep;
}, 40);

return () => clearInterval(interval);
...

```

A1.2.6 Rendering Colours

```

const formattedEntityColour = `rgb(${entityColour.r}, ${entityColour.g}, ${entityColour.b})`;
const formattedBackgroundColour = `rgb(.....)`;
...
<View style={{ backgroundColor: formattedBackgroundColour }}>
  ...
  <Circle fill={formattedEntityColour} />
  <Rect fill={formattedEntityColour} />
</View>

```